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ResQ: A hybrid classical-quantum model for efficient breast cancer image classification

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ABSTRACT

Breast cancer remains one of the leading causes of mortality worldwide, necessitating accurate and efficient diagnostic systems to improve treatment outcomes. While deep learning-based Computer-Aided Diagnostic (CAD) tools have demonstrated promise in analyzing histopathological images, they face challenges in handling high-dimensional data and computational inefficiencies. Simultaneously, quantum computing has emerged as a transformative technology, offering unparalleled capabilities in modeling complex data distributions and accelerating computations. This paper introduces ResQ, a hybrid classical-quantum framework designed for breast cancer classification. ResQ integrates a ResNet-based feature extraction module with a Variational Quantum Circuit (VQC) classifier, leveraging the complementary strengths of classical deep learning and quantum computing. Evaluations on two publicly available datasets, BreakHis and Bioimaging (BI), reveal significant performance improvements, achieving accuracies of 98.49% and 88.96%, respectively, compared to 97.36% and 87.96% achieved by its classical counterparts. Quantum circuit evaluations have been conducted on a quantum simulator (Aer simulator) as well as a noise-affected real quantum processor (*ibm_brisbane*). Additionally, to gain deeper insights into the effectiveness of the ResQ model over classical models, a non-parametric statistical test, viz., the Friedman test, followed by the Nemenyi test for post hoc analysis, is performed. Furthermore, a detailed circuit-level analysis explores critical trade-offs, such as circuit depth, gate count, and qubit usage, providing unique insights into the practical deployment of hybrid classical-quantum models in medical imaging. The one-qubit shallow architecture of ResQ renders lower circuit complexities, making it amenable to Noisy-Intermediate Scale Quantum (NISQ) devices. These findings underscore the potential of quantum computing to revolutionize cancer diagnostics by enhancing both accuracy and computational efficiency. The relevant codes of the proposed architecture, ResQ, are publicly available on https://github.com/DVLP-CMATERJU/ResQ-Hybrid-Classical-Quantum.

1. Introduction

Among the diseases causing the most casualties worldwide, cancer is in a leading position, affecting millions of people every year — not only impacting public health on a large scale but also posing financial and economic tolls on individuals and healthcare systems. Of the various types of cancer, breast cancer is particularly prevalent, posing significant health challenges for women worldwide [1,2]. Early detection and diagnosis play a crucial role in the timely management and cure of breast cancer. Medical imaging plays a pivotal role in identifying breast cancer, but it often relies on expert interpretation, which can introduce variability and human error. To address these issues, Computer-Aided Diagnostic (CAD) tools have become crucial, leveraging advancements in Artificial Intelligence (AI) to assist in distinguishing between benign and malignant tumors. Among the AI techniques, deep learning, and in particular Convolutional Neural Networks (CNNs), have shown significant promise for analyzing breast histopathology images owing to their exceptional capabilities in pattern recognition tasks by identifying discriminating features [3–5].

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Limitations of deep learning tools become evident when processing large-scale and high-dimensional low volume medical datasets with limited annotations characterized by complex data distributions, creating challenges in the scalability and efficiency of CAD tools [6–9]. Thus, memory limitations and processing delays in developing deep learning-based CAD systems in clinical environments serve as primary motivation to look beyond conventional deep learning techniques for cancer detection, necessitating innovative approaches.

Quantum computing has been introduced as a transformative technology that leverages the principles of superposition, entanglement and tunneling, enabling it to perform certain types of computations exponentially faster than classical systems. These principles allow quantum systems to efficiently model intricate data distributions and uncover patterns that may remain elusive by classical methods [10–12]. This makes it particularly suitable for tasks such as identifying subtle patterns and accelerating the processing of large datasets [13–17]. Organizations like IBM®, Google®, and Microsoft® have driven rapid advancements in quantum computing, evolving from systems with a few qubits to technologies capable of executing increasingly complex operations. Although quantum systems remain prone to noise and instability, ongoing research studies are addressing these challenges through improved algorithms and hardware robustness [18–23]. Despite this progress, the practical adoption of purely quantum CAD systems faces significant hurdles.

One of the primary challenges of using quantum computing is encoding classical data into quantum states. Furthermore, the lack of practical quantum devices for storing and processing large medical datasets remains a major bottleneck. Quantum RAMs, which are essential for handling quantum-encoded data, are implemented using quantum circuits that are susceptible to noise and errors [24–26] like other unconventional paradigms such as in Stochastic Computing [27–31], Approximate Computing [32]. This makes purely quantum deep learning models impractical at present. To address these limitations, hybrid classical-quantum systems have emerged as a promising solution.

Hybrid classical-quantum models combine the strengths of classical computing and quantum algorithms, enabling effective utilization of quantum resources while mitigating their current limitations. Parameterized Quantum Circuits (PQCs) are particularly noteworthy in this context, offering noise robustness and seamless integration with classical frameworks [33–36]. These hybrid models are capable of addressing scalability issues and improving circuit efficiency, making them well-suited for applications in medical imaging. For example, a hybrid approach can utilize classical deep learning methods for feature extraction while employing quantum algorithms for optimization and classification tasks, thereby leveraging the best of both paradigms. This paper introduces ResQ, a new hybrid classical-quantum framework for the classification of histopathological breast cancer images.

The motivations for the proposal can be summarized as follows.

- Necessity to deal with complex, high-dimensional histopathological data.
- Computational bottlenecks in terms of high GPU runtime, memory consumption, and inference latency, especially during deep model training and inference on low number of medical images, suffered by existing CAD tools.
- Ability of hybrid classical-quantum technologies to supersede purely classical ones in successfully modeling medical data.
- Leveraging Parameterized Quantum Circuits (PQCs) known for their noise-robustness.

ResQ leverages the complementary strengths of classical deep learning and quantum computing by integrating a ResNet-based [37] feature extraction module with a VQC classifier. ResQ is designed to overcome the limitations of traditional methods while maintaining model compactness. Evaluations on two publicly available datasets, BreakHis [38] and BI,[1] reveal that ResQ achieves significant performance improvements in accuracy, precision, and recall compared to classical CNNs. Additionally, a detailed circuit-level analysis explores trade-offs such as circuit depth, gate count, and qubit usage, providing unique insights into the potential of quantum technologies to enhance diagnostic accuracy and accelerate computational processes.

In summary, the contributions of this paper are as follows.

- The paper introduces a new hybrid network, ResQ, combining pretrained ResNet with a VQC classifier for breast cancer classification in histopathology images.
- The proposed model is tested on two publicly available datasets (BreakHis and BI) considering cross validations and standard deviations, achieving significant performance improvements (up to 10%) over classical CNNs.
- The paper compares the hybrid model with state-of-the-art methods, highlighting better accuracy and efficiency, with a detailed analysis of circuit depth, gate count, and qubit usage.
- Validity of the results obtained through quantum simulation has been re-established through execution on real quantum hardware that involves noise.

The rest of the paper is organized as follows. Section 2 discusses recent related works in the field. Section 3 outlines the foundational concepts, including Convolutional Neural Networks (CNNs), Transfer Learning, and Quantum Neural Networks (QNNs), and the proposed hybrid classical-quantum network. Section 4 details the experimental setup and procedures. Section 5 presents the results of the experiments and compares them with existing methods. Finally, Section 6 concludes with a discussion of key findings and potential future directions.

2. Related works

Machine learning and deep learning have emerged strong as ever in the domain of classification and regression problems, in recent years [39,40]. Quantum mechanics, applying its transformative potential over varied fields, shows remarkable prospects in the field of both machine learning and deep learning [41]. It provides a major advantage to machine learning and deep learning theories through innate parallelisms enabled by the principles of superposition and entanglement.

2.1. Quantum machine learning

Quantum machine learning refers to the quantum version of machine learning algorithms such as Support Vector Machines (SVM), k-Nearest Neighbors (KNN), and their likes. Instances of proposing Quantum Support Vector Machines (QSVMs) have been found in [42–46]. Delilbasic et al. proposed a QSVM [44] that executed pixel-based classification on the SemCity Toulouse dataset [47], identifying whether a pixel in an image belonged to a building or not. Quantum annealing [48] was used to train an SVM employing quadratic unconstrained binary optimization [49]. QSVM was employed by Li et al. to execute differentiation between the digits 6 and 9 [45]. The authors demonstrated exponential speedup over the classical SVM. A comparative study was carried out by Rana et al. [46] between classical and quantum SVMs for recognizing handwritten characters. Principal Component Analysis (PCA) technique using quantum model was proposed by Ostaszewski et al. [50] for classifying images. Quantum parallelization in distance calculations among observed instances was brought about by a Quantum KNN (QKNN). Ruan et al. [51] pioneered QKNN in measuring Hamming distance between feature vector sets of test and train images, stored as quantum states. Their work was extended over more complex datasets by Dang et al. [52] by extracting color features. Further improvements on QKNN were accomplished in the works of Wang et al. [53] for handwritten digit identification and Zhou et al. [54], for extracting features from images.

[1] http://www.bioimaging2015.ineb.up.pt/dataset.html.

2.2. Hybrid quantum neural networks

Hybrid Quantum Neural Networks (HQNNs) comprise frameworks that involve classical-quantum techniques clubbed together, where the quantum part requires shallow variational circuits [55–60]. In this context, a class of classical-quantum hybrid algorithms, known as variational quantum algorithms, deserves special mention for supervised learning [33,55,61–63]. Farhi et al. introduced a neural network based on quantum phenomenon focusing on classification of qubit-encoded discrete data involving an exponentially larger number of qubits [61]. In [64], a low-depth circuit to build a quantum algorithm for neural networks has been described. Schuld et al. designed a quantum-inspired slim machine learning model circuit [65]. In 2019, Henderson et al. introduced a type of transformational layer similar to classical convolutional filter layers – consisting of a number of random quantum circuits – called a quantum convolution, or quanvolutional layer [66]. Operating on input data, the work aimed at producing meaningful features for the classification of images. Trochun et al. [67] introduced a hybrid CNN comprising three convolution and two fully connected layers connected to a VQC, for detecting hurricane damage. Another work by Sebastianelli et al. [68] performed classification of remote sensing images with a simple hybrid network. Chalumuri et al. [69] used real quantum computers of up to 9 qubits to emulate a hybrid network consisting of a quantum feature extraction phase and a classical deep neural network to perform image scene classification. It was demonstrated that the quantum layers could outperform state-of-the-art results using one-third parameters. Guha et al. [60] proposed an ensemble framework combining three hybrid classical-quantum neural network architectures that demonstrated reasonable accuracy in breast cancer detection from histopathology images. Binary classification tasks exploring Variational Quantum Deep Neural Network (VQDNN) classifiers [70], Barren Plateaus phenomenon [71], Minimum Redundancy Maximum Relevance (mRMR) [72], and Principal Component Analysis (PCA) have been found to be rampant in the literature. Each of the mentioned phenomena utilized hybrid systems. In comparison with CNNs, the Quantum Convolutional Neural Networks (QCNNs) presented higher classification accuracy as well as faster training capabilities.

Table 1 summarizes the recent works worth taking a note of, in the domain of quantum machine learning, quantum deep learning, and hybrid classical-quantum integration architectures, highlighting the working principle of the models, novelty of the proposal, merits, and demerits.

In [58], authors presented a hybrid classical-quantum transfer learning framework for classifying mammogram images as malignant or benign. A pretrained ResNet-18 classical CNN was used to extract 512 features from the input images, which were passed through a VQC consisting of a classical-to-quantum angle embedding, trainable variational layers with entanglement, followed by measurement to produce final predictions. For choosing ResNet18, they compared results across 5 different architectures (AlexNet, VGG-19, DenseNet-161, ResNet-18, and ResNeXt-50 32 × 4d) using heatmaps resulting from GradCAM and GradCAM++ that were used to interpret CNN focus area. Depth sizes from 1 to 4 were considered for the quantum circuit. The dressed quantum circuit with depth = 1 — CQ1 was tested on a real quantum device – ibm_lagos – using the best model found by the quantum simulator (where the best F1-score was measured). Twenty repetitions of the same testing showed dissimilar results across repetitions, but the model was found to be stable.

In [73], Majumdar et al. proposed a hybrid classical-quantum deep learning architecture to classify histopathological images from the PatchCamelyon (PCam) dataset into cancerous and non-cancerous categories. The authors analyzed ResNet-18, VGG-16, Inception-v3, and AlexNet transfer learning models. The final fully connected layer of these models was modified with a QNN. Repeated VQC layers that had been tested for 6 configurations of single-qubit rotational and two-qubit controlled gates were considered to build the VQC-based QNN. The best VQC with the highest expressibility was finalized. A fully connected neural network — classical Artificial Neural Network (ANN) was the chosen last unit of the presented architecture to which the output of the quantum unit (VQC-based QNN) was mapped, with a softmax activation function. The classical ResNet-18 outperformed all other models in terms of both accuracy and AUC. In [74], the authors proposed a method, CVFBJTL-BCD, that combined multi-modality medical imaging, transfer learning, and deep learning to diagnose breast cancer. The authors employed the Gabor Filtering (GF) technique for noise removal, chosen for its effectiveness in conserving significant spatial frequency data while reducing unwanted noises. Features extracted from the joint transfer learning process of three pre-trained CNNs (DenseNet-201, Inception-V3, MobileNet-V2), were fused. The Stacked Autoencoder (SAE) technique was employed to precisely classify breast cancer. The Horse Herd Optimization Algorithm (HHOA) model was used to tune parameters for the SAE method. The proposed model showed 98.18% accuracy for the BreakHis dataset as well as high percentages of precision, sensitivity, and specificity. On the ultrasound data, a test accuracy percentage of 99.15 was obtained.

Rana et al. [75] presented a hybrid QCNN framework for the automated diagnosis of Tuberculosis (TB) from Chest X-rays (CXR). Data preprocessing and postprocessing were handled on classical machines while the quantum circuits were utilized for the core computational tasks. Novel Enhanced Quantum Representation (NEQR) was used for encoding classical data into quantum states. The penultimate layer in the classical neural network was replaced with a quantum layer built from a VQC. The output of the VQC was then used to make a prediction. Incorporation of hybrid QCNN showed validation accuracy of 87% during training. However, on the test set, the accuracy percentage fell short by 0.01% than the classical Xception, ResNet-50, and VGG-16 models. Hybrid QCNN yielded better performance metrics measured in terms of precision, sensitivity, and F1-score than those of Xception, ResNet-50, and VGG-16.

Das et al. [76] introduced Shallow Quantum Neural Networks (SQNNs) as hybrid models that combined classical deep learning architectures (ResNet and VGG-16) with PQCs to enhance image classification tasks—specifically, the identification of surface cracks in concrete images. Two types of quantum networks were developed — CircuitQNN and TwolayerQNN. These were incorporated into ResNet and VGG-16 models to build ResCQ-SQNNs, ResTQ-SQNNs, VGG16VCQ-SQNNs, and VGG16VTQ-SQNNs. The approach considered the use of a classical neural network and added a few quantum layers governed by the mentioned pre-set quantum circuits. When the sample size is large, both ResNet and VGG-16 outperformed the corresponding SQNNs. Only for a smaller subset of data, ResCQ, ResTQ, VGG16CQ, and VGG16TQ showed better performance.

The first-of-its-kind multi-modal data fusion combined with quantum neural network "Quantum Neural network-based Multimodal Fusion system for intelligent diagnosis (QNMF)", was proposed by Qu et al. [77]. The system had been designed to perform intelligent medical diagnosis by fusing data from various sources (e.g., medical images and Electronic Health Records — EHRs). Medical features extracted using QCNN-based feature extractor were fused with other modality features to train a VQC for intelligent diagnosis of breast cancer and COVID-19. Internet of Medical Things (IoMT) devices transmitted multimodal medical data required for this QCNN-based multimodal fusion system. QCNN was optimized using a classical backpropagation algorithm and exhibited robustness against quantum noise. In COVID-19 diagnosis, the quantum fusion model showed 97.61% accuracy, better than the accuracies obtained by single modalities like CT scan images and blood test samples fed to QCNN and VQC models. For breast cancer data, QCNN with ultrasound images with 83.86% accuracy, fell way short of VQC with cell slices features at 95.32% accuracy. However, the best accuracy percentage was shown by the quantum fusion model at 97.07%.

Table 1
Critical comparison of a few notable works in the hybrid classical-quantum domain along with the proposed work.

Ref.	Quantum classifier and backbone	Methodology and dataset	Performance	Novelty/Merits	Demerits
[58]	VQCs of several depths. **Backbone:** Pretrained ResNet18 with VQC of depth 1 to 4.	Used a hybrid classical-quantum transfer learning framework to classify mammogram images as malignant or benign. **Dataset:** Breast cancer digital repository (BCDR), mammogram image dataset with 2 classes.	Non-quantum pre-trained ResNet18 using transfer learning and two hybrid VQCs having depth 1 and 4 reported best accuracy values. VQC of depth 1 yielded precision and specificity of 1.00. AUC values of the hybrid models were better than both the pre-trained and non-pre-trained classical models.	1. For CNN selection, heatmaps generated from GradCAM and GradCAM++ led to DenseNet-161, ResNet18, and Res-NeXt-50 32 × 4d initially for training and testing on a quantum simulator. 2. Full mammogram classification using a quantum classifier. 3. Usage of circuits of varying depth (1–4) for optimal performance was the first of its kind.	1. No statistical analyses. 2. Only single datasets were examined. 3. Only 825 mammograms involving high class imbalance were considered. Data augmentation would have led to bias introduction. 4. VQC complexity was not reported. 5. Low recall vis-a-vis high specificity indicated missing malignant cases with more conservative models. 6. No reference to computational times. 7. Codes are not publicly available.
[73]	Several (VQCs) of different configurations. **Backbone:** Multiple transfer learning models - ResNet-18, VGG-16, Inception-v3, AlexNet with VQC.	Proposed a hybrid classical-quantum deep learning architecture to classify histopathological images as cancerous and non-cancerous. The final fully connected layer of DL models was modified with a QNN. **Dataset:** 10k out of 100k images from PatchCamelyon (PCam) — digital pathology images	Hybrid ResNet18 + VQC achieved 88.5% accuracy and 0.93 AUC with 1k images, outperforming some classical CNN baselines. However, when trained on full 10k images, hybrid models fell slightly behind classical models — 82.35% vs. 89.9%.	Authors generated 6 variational quantum circuits of varied configurations.	1. No noise simulations or execution on real noisy quantum hardware. 2. Other CNN architectures apart from ResNet-18, VGG-16, Inception-v3, and AlexNet, have not been defined. 3. Use of 4, 5, 7 qubits for encoding classical CNN outcomes resulted in complex VQCs limiting performance compared to the ResNet-18 model. 4. Codes are not publicly available.
[74]	No quantum layer. **Backbone:** DenseNet-201, Inception-V3, MobileNet-V2.	Combined multi-modal medical imaging, transfer learning, and deep learning to diagnose breast cancer. **Dataset:** Multi modal medical images — histopathological datasets (BreakHis), Breast Ultrasound dataset.	Showed 98.18% accuracy for the BreakHis dataset as well as high percentages of precision, sensitivity and specificity. On the ultrasound data, test accuracy of 99.15% was obtained.	A novel fusion-based joint transfer learning paradigm integrating multiple pretrained CNN architectures was considered. For edge-preserving noise reduction, the uncommon use of Gabor filters was done before diving into deep learning.	1. Did not encompass any quantum advantage rendering it a non-quantum work in the classical domain. 2. The BreakHis data was divided into a $7:3$ training and testing ratio with no validation. 3. Codes are not publicly available.
[75]	NEQR image encoding followed by feature extraction and model optimization as part of QCNN. **Backbone:** Pre-trained Inception, Xception, ResNet50, and VGG-16.	Presented a hybrid QCNN framework for the automated diagnosis of TB from chest X-rays (CXR). **Dataset:** Two publicly available chest radiograph datasets from Himachal Pradesh, India.	QCNN showed validation accuracy of 87%. Test set accuracy fell short by 0.01% over the classical Xception, ResNet-50, VGG-16 models. Precision, Sensitivity, F1-score of QCNN were better over those of Xception, ResNet-50, and VGG-16.	Included rigorous statistical testing (e.g., t-tests and confidence intervals) to confirm performance improvements.	1. A comparatively smaller dataset (180 CXR images) was used. 2. Three classical models outperformed the QCNN in terms of accuracy, recall, and F1-score percentages. 3. The QCNN was evaluated using quantum simulators, and not on real quantum hardware, misrepresenting practical performance. 4. Codes are not publicly available.
[76]	Shallow QNN (SQNN) involving — Circuit-QNN, TwoLayer-QNN. **Backbone:** ResNet, VGG-16	Introduced SQNNs as hybrid models that combine classical deep learning architectures (ResNet and VGG-16) with PQCs forming CircuitQNN and Two-layer QNN to perform surface crack identification. **Dataset:** "Surface Crack Detection" set available on Kaggle.	Both ResNet and VGG-16 outperformed the corresponding SQNNs for large sample sizes. For smaller susbset of data, ResCQ, ResTQ, VGG16CQ, VGG16TQ showed better performance.	SQNN architectures have not been studied previously on image processing tasks specifically for crack detection.	1. Experiments were restricted to smaller datasets (5% of full size) due to the small number of qubits available in IBM simulators. 2. Simulations were done using the IBM Aer simulator and not on real noisy quantum hardware, potentially misrepresenting practical performance. 3. No statistical analyses were reported. 4. Codes are not publicly available.

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Table 1 (*continued*).

[77]	QCNN and VQC with data from multiple sources.	Proposed QNMF, a Quantum Neural Network-based Multimodal Fusion system designed to perform intelligent medical diagnosis (breast cancer and Covid-19) by fusing data from various sources. **Dataset:** 1. Medmnist, Breast Cancer Wisconsin. 2. Covid-19 CT scan image dataset, Covid-19 blood test features.	In Covid-19 diagnosis, the quantum fusion model showed 97.61% accuracy. For breast cancer data, QCNN with ultrasound images with 83.86% accuracy, fell way short of VQC with cell slices features at 95.32% accuracy. QNMF showed 97.07% accuracy.	1. One of the first frameworks combining quantum computing with multimodal data fusion in healthcare, surpassing traditional unimodal Quantum Machine Learning (QML) applications. 2. 4 sources of quantum noise have been considered for resilience analysis. 3. Comprehensive mathematical proof of adapting the classical backpropagation algorithm to one suitable for QCNN has been presented. 4. Reproducibility and experimentation are ensured due to publicly available code and datasets.	1. Classical pre-trained neural network models were compared but no pre-trained classical model was combined with QCNN or QNMF even though the classical models displayed promising results over the quantum methods at multiple instances. 2. The system was only evaluated using quantum simulators (Pennylane), not on real quantum hardware. 3. Only small input sizes (e.g., 32×32 images) were used due to simulator limitations, leaving the architecture's scalability to complex tasks and high-resolution medical images unproven.
[78]	Amplitude Encoding-Variational Quantum circuit (AE-VQC). **Backbone:** ResNet and MaxViT	Introduced a hybrid classical-quantum image classification framework that replaced the traditional global pooling layer in deep learning models with a VQC. **Dataset:** Croatian fish dataset, aircraft dataset, breast ultrasound images, Apples or Tomatoes (AOT).	ResNet-18_q exhibited improvements in test accuracy of up to 5.70% and in test F1-score of up to 5.53%, over the classical ResNet-18 model. MaxViT-T_q provided up to 9.21% higher accuracy and up to 15.79% higher F1-score, when compared with MaxViT-T.	1. Variational Quantum Algorithms (VQAs) were employed for global pooling for the image classification task instead of using pooling with quantum analogues. 2. Training set epoch execution time was reported. 3. Demonstrated that quantum circuits can handle classification with fewer parameters while maintaining or improving performance.	1. A 15-qubit ansatz encoded the feature maps from the final layers of ResNet-18. This large number of qubits prevented the training of the architecture using gradient-based optimizers, which otherwise would inculcate a huge number of complex circuit executions. 2. Highly volatile qubits in real quantum hardware may render this architecture susceptible to noise, leading to performance degradation. 3. Amplitude encoding used, resulted in an exponential increase in quantum circuit depth. 4. Experiments were performed on simulators and not on real quantum hardware, misrepresenting practical performance. 5. No statistical analyzes were reported. 6. Codes are not publicly available.
[79]	Quantum support vector classifier **Backbone:** MobileNet pre-trained network	Explored integration of quanvolutional neural network and Quantum Support Vector Classifier (QSVC) with classical deep learning models for classification of skin lesions. **Dataset:** HAM10000 dermatological image dataset.	Combining RY qubit rotation with the Pauli-Z gate in the proposed network produced the highest classification accuracy — 82.86%, whereas without pre-training, QSVC produced an accuracy of 72.5%.	1. Implemented and compared both quanvolutional neural networks and QSVMs. 2. Evaluated multiple combinations of qubit rotations (RX, RY, RZ) and Pauli gates (X, Y, Z) in the quanvolutional layer to determine optimal configuration. 3. Measured performance across 7 skin lesion categories with confusion matrices.	1. Reduced image dimensions at the very onset of the processing pipeline, may result in the exclusion of important features. 2. Did not compare with any recent state-of-the-art method. 3. Lack of algorithmic and mathematical explanations rendered the methods vague and incomprehensive. 4. Statistical analysis as well as implementations on real quantum hardware are missing.

(*continued on next page*)

Table 1 (*continued*).

[14]	Variational Quantum Classifier. **Backbone:** ResNet-50	A hybrid model, the QuantumNet was built by integrating ResNet-50 with a 4-qubit VQC for diabetic retinopathy (DR) detection. **Dataset:** APTOS 2019 blindness detection dataset.	QuantumNet achieved 94.11% accuracy, outperforming ResNet (90.20%), CNN (85.14%), and MobileNet (82.68%).	1. The hybrid design was first-of-its-kind integration of ResNet-50 with a VQC for DR detection. 2. QuantumNet was evaluated against CNN, MobileNet, and ResNet using the APTOS 2019 dataset with statistically validated results (Wilcoxon signed-rank test).	1. No implementation on real quantum hardware. For a 4-qubit circuit, quantum hardware may exhibit larger operational errors and decoherence drastically, changing the outcomes.
Proposed	Variational Quantum Classifier (VQC) **Backbone:** ResNet-18, ResNet-34, ResNet-50, ResNet-101, ResNet-152	A hybrid classical-quantum ResQ model, involving a single-qubit VQC, is designed for binary classification of breast cancer in histopathological images. **Dataset:** Breast Cancer Histopathological Database (BreakHis), BI Database.	Outperforms classical ResNet models on both datasets. For BreakHis dataset, accuracy up to 99.83% is obtained and for the BI dataset accuracy up to 88.96% is obtained.	1. A VQC is used without entanglement, which often leads to equally good or even better model performance [80]. 2. Benchmarking on real quantum processors carried out. 3. Extensive statistical analysis — Friedman Test and Nemenyi Test carried out. 4. Performing a first-of-its-kind circuit-level comparison between ResQ and other classical-quantum hybrid models. 5. Training set execution times for both classical and hybrid networks reported. 6. Execution of similar work and performance trade-offs discussed. 7. Time complexity analysis showcased.	1. Using only one qubit may limit model's capacity and expressiveness.

Chen et al. proposed a hybrid classical-quantum image classification framework that replaces the traditional global pooling layer in deep learning models with a VQC using amplitude encoding [78]. The feature maps from classical CNN backbones (e.g., ResNet, MaxViT) were encoded into quantum states using amplitude encoding, which were then processed through two different PQCs that transformed the quantum states. The decoding comprised observing the expectation values obtained by measuring the Pauli-Z operators and passing them to the classical layers. Two quantum simulators — a state vector simulator ($SCNN_q_sv$) and a shot-based simulator ($SCNN_q_sb$) were employed to train and test the model. $ResNet-18_q$ exhibited improvements in test accuracy and F1-score of up to 5.70% and 5.53% respectively, over the classical ResNet-18 model. $MaxViT-T_q$ yielded up to 9.21% higher accuracy and up to 15.79% higher F1-score, when compared with MaxViT-T. However, the time elapsed to train the quantum models has been found to be more than that required to train the non-quantum models.

Reka et al. explored the integration of QML with classical deep learning models for the classification of skin lesions using the HAM10000 dermatological image dataset [79]. Two quantum approaches — Quanvolutional Neural Network and Quantum Support Vector Classifier (QSVC) were used to segregate skin lesion images into seven different classes. For the QNN, regular data augmentation methods like rotation, flipping, and contrast enhancement were randomly applied to the original dataset to mitigate class imbalances. Pre-trained MobileNet was used for feature extraction, followed by min–max normalization. Pre-processed, normalized images were resized to 28×28. Then again, a small region of size 2×2 was encoded to a 4-qubit quantum circuit accomplished by parameterized rotations of RX, RY, and RZ with angles scaled by a factor of π. The QSVC comprised Hadamard gates for superposition, ZZ feature map for quantum feature extraction, and FidelityQuantumKernel for multi-classification. Pauli-X, Pauli-Y, and Pauli-Z gates were used for the measurement of expectation values. The quantum convolution was succeeded by classical layers and a SoftMax activation function at the classification layer. The mentioned steps were repeated across multiple image sections that allowed scanning the entire image, giving rise to a multi-channel output image. Combining RY qubit rotation with the Pauli-Z gate in the quanvolutional neural network produced the highest classification accuracy — 82.86%, whereas without the pre-training, Quantum Support Vector Classifier produced an accuracy of 72.5%.

Bali et al. [14] considered three classical deep learning models — CNN, ResNet, and MobileNet, and evaluated them for diabetic retinopathy detection. The best-performing model, ResNet-50, was eventually used for feature extraction and integrated with a 4-qubit VQC. The hybrid model called QuantumNet was evaluated through quantitative metrics such as accuracy, precision, recall, and F1-score, as well as validated statistically using techniques like the Wilcoxon rank test. Experiments were carried out on the Google Cirq quantum simulator. QuantumNet achieved 94.11% accuracy, outperforming ResNet (90.20%), CNN (85.14%), and MobileNet (82.68%).

The ResQ model proposed in our work is a first-of-its-kind wherein in-depth resource analysis encompassing quantitative metrics like depth, algebraic complexity, and gate count for measuring complexity of comparable quantum circuits, has been considered. ResQ is shown to be more resource-efficient, using fewer qubits, lower gate count, and shallower circuit depth compared to prior hybrid models. The parameter-shift rule [81,82] employed for evaluating quantum gradients has been well-explained with mathematical rigor. Robust experimental setup with diverse ResNet configurations (18, 34, 50, 101, 152) as backbones, achieved accuracies of up to 99.83% (BreakHis) and 88.96% (BI). Furthermore, the BreakHis dataset across magnifications was split in three different ways, encompassing random split, uniform

split, as well as patient-level split. The comprehensive empirical study, along with comparative performance, shows noticeable gains over the classical models and state-of-the-art classical-quantum hybrids. Time complexity analysis of the hybrid architecture is another aspect that is unprecedented in previous works on such architectures. Extensive statistical tests have been applied to verify the significance of performance differences across classifiers. Our work additionally engages a noise-affected real quantum processor to evaluate the performance of the ResQ configuration that attained the best results on a quantum simulator, as well as another hybrid classical-quantum architecture to establish time gains.

2.3. Quantum transfer learning

Transfer learning refers to the paradigm of transferring pre-acquired knowledge from one network to another. In the context of machine learning or deep learning, the transfer of knowledge occurs by applying pre-trained weights – those made available by training a specific network using a given set of data – to a separate network of interest [83]. This alleviates the requirement of training deep networks over large datasets, which is typical in computer vision problems. Transfer learning has been broadly resorted to in the literature [84].

This section is dedicated to providing a summary of the research work being undertaken in the paradigm of transfer learning in building quantum neural networks, with a focus on ResNet as the choice of classical convolutional neural network (CNN). In this regard, particular emphasis has been made on the "classical to quantum (CQ)" transfer learning variant. Classical to quantum transfer learning garners the advantage of exploiting the power of any state-of-the-art classical deep neural network as an extractor of highly informative features that can be subsequently manipulated using VQCs.

The pioneering research in this regard was the work done by Mari et al. in 2020 [56]. They outlined a first of its kind of framework where classical neural networks could be connected with VQCs to build hybrid computational models. Pre-trained ResNet-18 served as the classical feature extractor. Engaging both quantum simulators as well as real quantum devices, an image classification problem was addressed that showed promising results. Blankson and Pavlovskiy applied classical ResNet-18 as well as ResNet-18 combined with a continuous-variable photonic quantum model on a speaker classification dataset [85]. Apart from ResNet, the photonic quantum model was also combined with a simple convolutional neural network, which outperformed the ResNet-18 quantum model significantly after one epoch. An instance where features extracted from an image using ResNet-18 CNN are thereby classified utilizing VQC, is found in the work by Sridevi et al. [86], in 2022. Diagnosis of diabetic retinopathy from Optical Coherence Tomography (OCT) retinal images was the addressed task. Retaining only the significant features as indicated by diagnosis, the 4-bit reduced feature vector classified by the VQC showed better accuracy than the existing state-of-the-art methods. In 2023, yet another instance of using quantum-augmented ResNet-18 as the transfer learning model for pre-training was proposed by Soto-Paredes and Sulla-Torres in a study on classifying face images of individuals wearing COVID-19 masks [87]. The goal was to classify images of people wearing masks into categories of — (i) correctly worn mask, (ii) incorrectly worn mask, and (iii) not wearing a mask. Accuracy as high as 99.5% in classifying people wearing masks correctly was obtained by the model. Transfer-learned VGG-16 was combined with multi-qubit quantum machine learning layers by Otgonbaatar et al. [88] to demonstrate effective benchmarking and validation capabilities on real-life datasets of both low and high dimensionalities. In 2024, Kujani et al. proposed a hybrid quantum-based ResNet-18 model based on transfer learning to carry out categorization and diagnosis of diseases affecting maize plants [89]. The model outperformed its non-quantum version, giving higher percentages of accuracy, precision, recall, and F1-score. Apart from these, several works in the hybrid architecture domain discussed in the previous section too, rely on the transfer learning paradigm [14,73,75,76,79].

3. Methodology

This section provides an in-depth discussion of the techniques and frameworks employed in the proposed hybrid classical-quantum neural network for breast cancer classification.

A flowchart illustrating the outline of the entire workflow is depicted in Fig. 1.

3.1. Convolutional Neural Networks (CNNs)

Computer vision tasks have received strong impetus from the advent of CNNs built of layers of convolution, pooling, and fully connected layers, bringing about linear and non-linear transformations on data [37,90–92]. Non-linear transformations through these layers aid in extracting representative features of the data of interest. The convolution operation applies a filter (or kernel) over a defined window of the input image to detect spatial hierarchies and patterns. For a single filter matrix $F_{i,j}$ of size $m \times n$ applied to an input image matrix A with dimensions $a \times b$, the convolution is mathematically represented as

$$A_{x,y} = \sum_{i=1}^{m} \sum_{j=1}^{n} A_{x+i,y+j} \times F_{i,j}.$$

The resulting output matrix has dimensions reduced to $\frac{a-m+1}{s} \times \frac{b-n+1}{s}$, where s is the stride. Multi-channel images and multiple filters extend this operation into higher dimensions, increasing the depth of the output feature map. These transformations make CNNs particularly adept at feature extraction and classification tasks.

3.2. Transfer learning

Deep learning approaches are bottlenecked by the necessity of having to train huge amount of sample data. ImageNet [93] is one such database comprising millions of labeled images that aid in improving the precision of training a complex deep learning model built of hundreds of layers. However, in the medical domain, collection and procuring images from patients and annotating them to generate large datasets is not only an expensive task, but also infeasible at times. Transfer learning is a strategy that comes to the rescue in that it re-uses the learnings from a particular task in order to perform a similar task [94–96]. A pre-trained CNN on a generic natural image database, or a cross-domain medical database, is applied to initiate another network. The rest of the training may involve either all layers of the network or a subset of them, supervised on the basis of the new database for the task of concern. This scheme is classified as fine-tuning the deep CNN. Another scheme exists in transfer learning in which all layers (except the final fully connected layer) of the pre-trained CNN are frozen, rendering it as a fixed feature extractor wherein images from the new database are served as inputs.

Many instances in recent research have shown to achieve state-of-the-art results by exploiting transfer learning in place of training CNNs from scratch, mitigating the evils of the likes of overfitting, data scarcity, etc. For histopathological image classification, too, transfer learning has shown promising results in earlier works [97–99].

3.3. Quantum Neural Networks (QNNs)

Quantum phenomena are inherently non-linear in nature, thereby being well-suited for complex feature extraction tasks, typical of neural networks. A QNN is built using VQCs, which depend on input data x and trainable parameters θ. The mathematical representation of a VQC is given by

$$f(x, \theta) = \mathrm{tr}[O(x, \theta)\rho(x, \theta)],$$

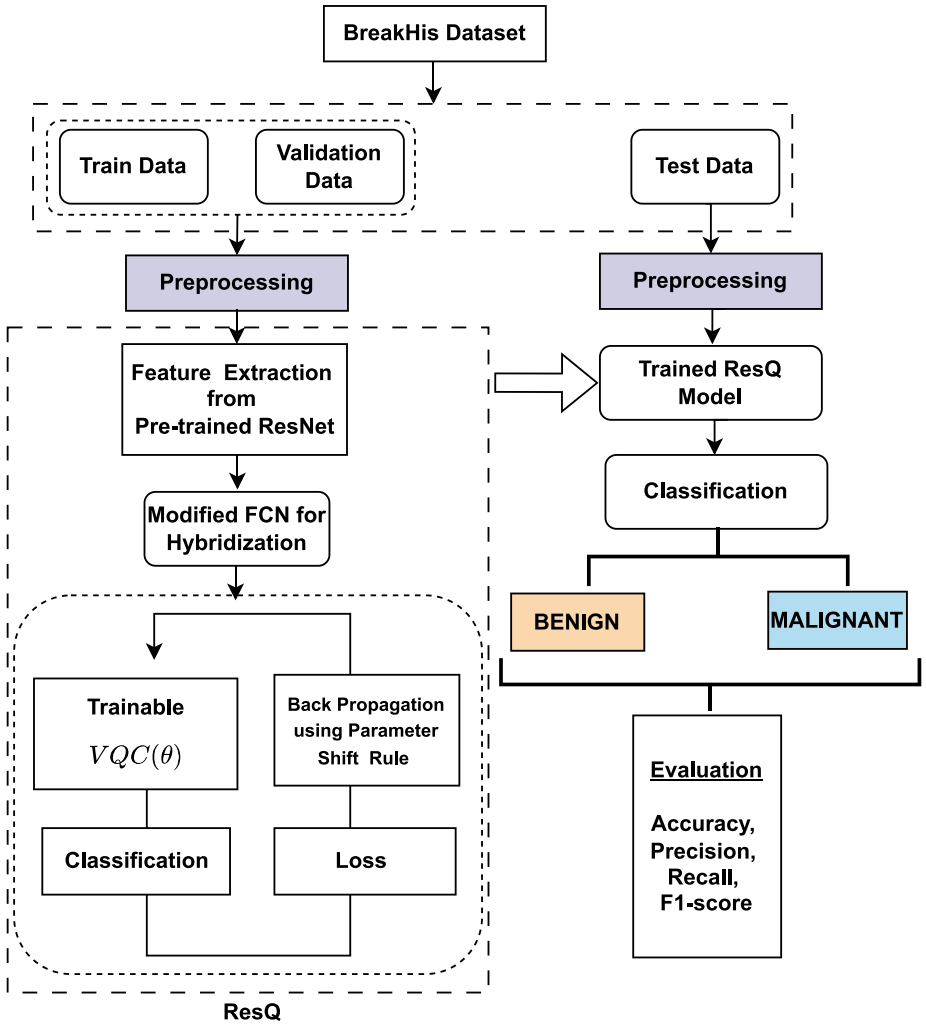


Fig. 1. Flowchart illustrating the principle of operation of the proposed ResQ architecture.

where, O is an observable, and ρ is the density matrix defined as a probabilistic mixture of quantum states

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|.$$

Like many other quantum versions of traditional methods, the mathematics of QNNs also often bears little or no resemblance to traditional perceptrons. Optimization techniques of such neural networks may be adapted from traditional optimizers [61] or may again involve mathematically dissimilar methodologies. Hybrid neural networks are those, whose framework comprises classical-quantum techniques clubbed together, wherein the quantum part requires shallow variational circuits [55–59]. The quantum layer comprises unitary operations, U on a set of inputs in the form of qubit states $|state_{inp}\rangle$. $|state_{inp}\rangle$, which passes through the shallow variational parameterized circuit to generate dimension-preserving output states $|state_{oup}\rangle$. Thus, the quantum layer can be represented as

$$\mathcal{QL} : |state_{inp}\rangle \rightarrow |state_{oup}\rangle = U(\theta)|state_{inp}\rangle, \tag{1}$$

where, θ is the parameter(s) which corresponds to *weights* in classical neural networks. Multiple such quantum layers may be concatenated together to build the variational quantum circuit (Q_C) of depth d as

$$Q_C = \mathcal{QL}_1 \cdot \mathcal{QL}_2 \cdot \mathcal{QL}_3 \ldots \mathcal{QL}_d. \tag{2}$$

An encoding layer $\mathcal{E}$ takes care of converting inputs from the classical vector space x to the quantum Hilbert space $|x\rangle$. A measurement layer $\mathcal{M}$ performs the reverse mapping $|x\rangle \rightarrow y$. In between, an approach that is popularly used for evaluating gradients for VQCs is the parameter-shift rule [81,82]. Alleviating the necessity of ancilla qubits, the parameter-shift rule requires the performance of simply two circuits, each of which has the same number of gates as the original circuit. The gradient of the loss function in the parameter-shift rule enjoys the advantage of being bounded by the number of parameters, thereby mitigating the problem of "exploding" gradients faced in stochastic gradient descent in classical machine learning. Eventually, the end-to-end network is essentially a mapping from classical to classical vector space guided by classical weights. The quantum layer behaves like a black-box - just like the way a classical deep neural network does in the case of a traditional CNN.

3.4. ResQ: a novel hybrid classical-quantum neural network

In this work, a hybrid classical-quantum neural network, ResQ, for binary image classification has been introduced. The proposed model combines the strengths of classical deep learning (i.e. ResNet) for feature extraction with the quantum computational power of a VQC. The framework consists of two primary components: a classical feature extraction module and a quantum classification layer. Each component is detailed below. The overall architecture is illustrated in Fig. 2, where ResNet-L indicates transfer-learned classical ResNet architectures for L = 18, 34, 50, 101, 152 and ResQ-L stands for the corresponding ResQ architectures.

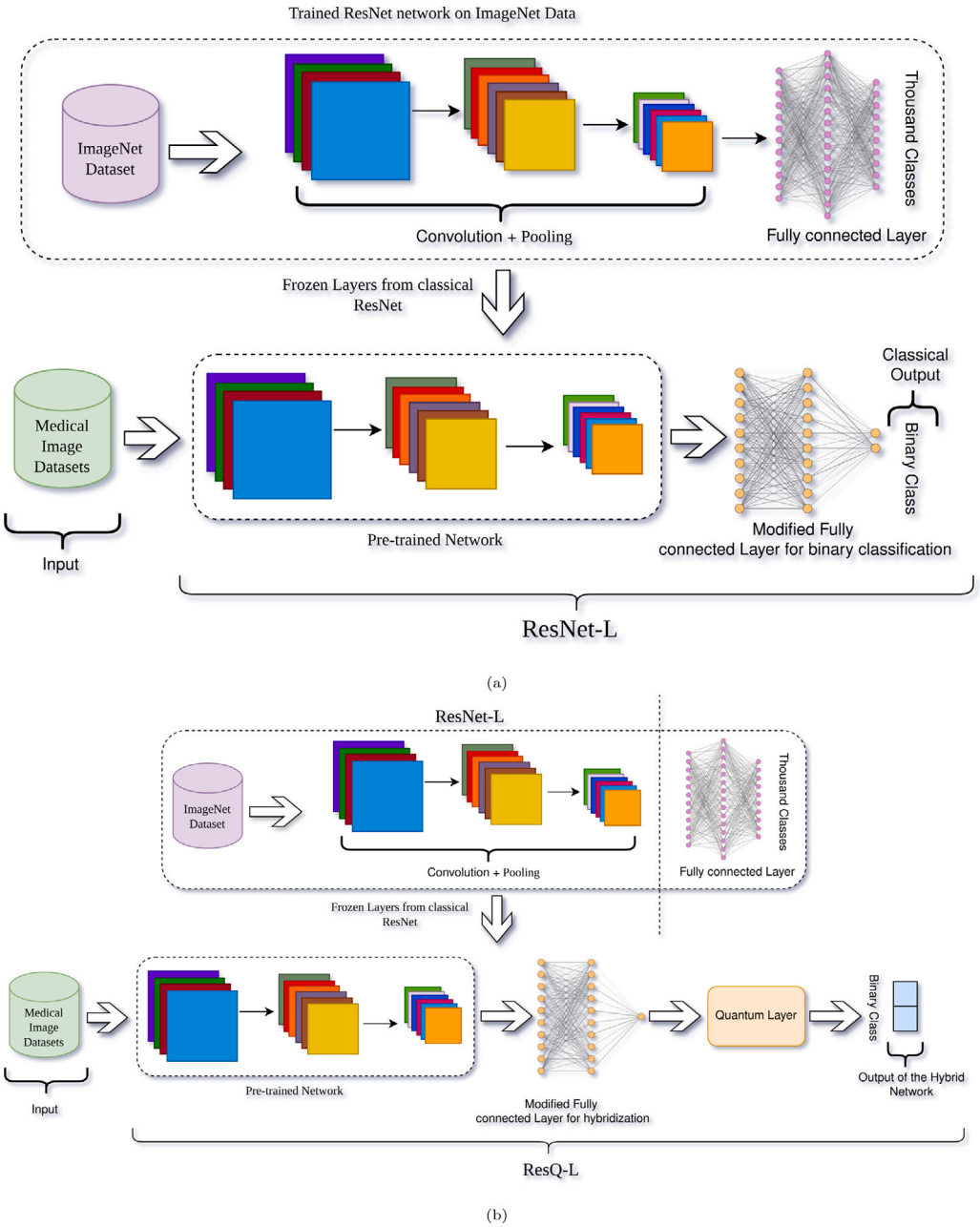


Fig. 2. Flow diagram of the network architecture depicting (a) Transfer-learned classical ResNet-L (b) The proposed hybrid network ResQ-L (L = 18, 34, 50, 101, 152).

3.4.1. Classical feature extraction module

The classical feature extraction module leverages the power of CNNs to extract meaningful features from input images. CNNs are highly efficient in capturing spatial hierarchies and patterns in visual data. For experimentation, multiple pretrained CNN architectures were evaluated, including VGGNet [100], DenseNet [101], ResNet [37], and Inception [102], pre-trained on the ImageNet dataset [93]. Performance was assessed based on metrics such as accuracy, precision, recall, and F1-score (Fig. 3). Among these, ResNet demonstrated superior performance due to its robust learning capabilities, fast convergence, and ability to address the vanishing gradient problem [103–105]. Based on these findings, variants of ResNet are adopted as the primary feature extractor for this work. Since the pre-trained ResNet generates a 1000-dimensional output in its final layer (corresponding to ImageNet classes), modifications were required to align with the input requirements of the quantum classification layer. Specifically, the final fully connected layer of ResNet is replaced with a custom configuration. This configuration includes: a linear layer, a ReLU activation applied to introduce non-linearity, and another linear layer that reduces the output dimension to a single scalar feature, which serves as the input for the quantum classifier.

3.4.2. Quantum classification layer

The quantum classification layer comprises a VQC parameterized by the output of the previous classical layer. For this, it is necessary to perform classical-to-quantum encoding that produces quantum states. The encoding of choice in this work is the one that produces a balanced superposition of the qubit states $|0\rangle$ and $|1\rangle$. This is accomplished by initializing the circuit to operate on a single qubit initialized in the $|0\rangle$ state. The state $|0\rangle$ then undergoes quantum transformations through a Hadamard gate (H gate), which places the qubit into the desired balanced superposition. Thus, the encoding layer $\mathcal{E}$ does the following

$$\mathcal{E}(\theta) = \mathcal{F}(\theta) H |0\rangle, \tag{3}$$

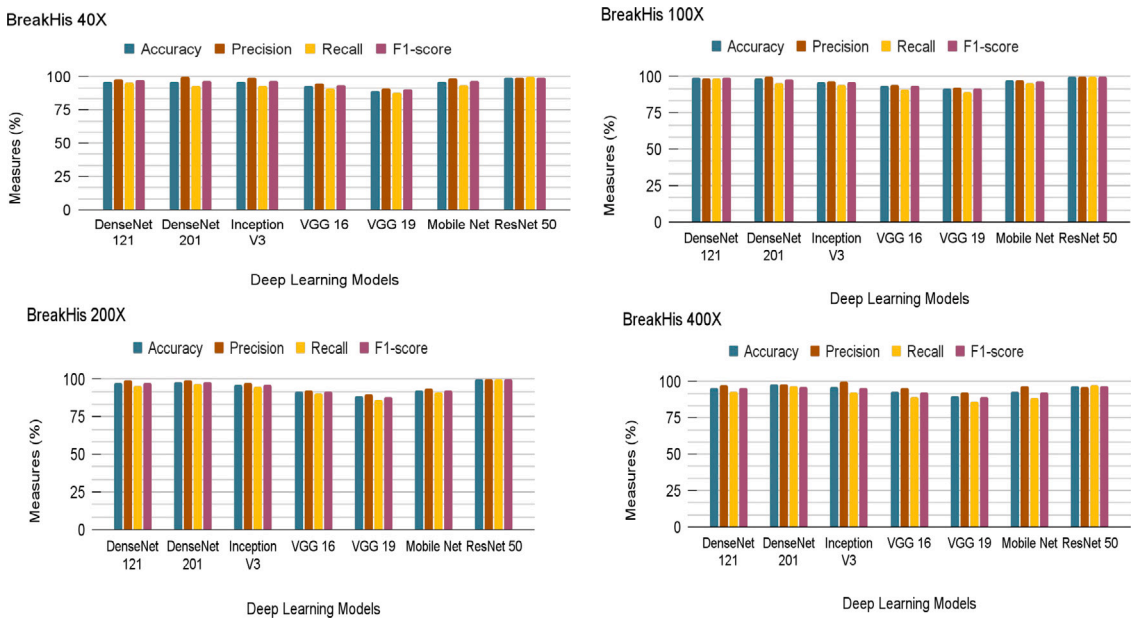


Fig. 3. Performance comparison of transfer-learned DenseNet-121, DenseNet-201, Inception V3, VGG-16, VGG-19, MobileNet, and ResNet-50 in terms of accuracy, precision, recall, F1-scores (expressed in percentages) of hybrid models on the BreakHis dataset for all four magnification factors.

where, $\mathcal{F}(\theta)$ is actually the parameterized gate, θ being the classical vector which serves as the parameter. A parameterized rotation gate $R_y(\theta)$ is chosen for the purpose, where the parameter θ is derived from the scalar feature output of the classical feature extractor. The feature output being a real number, the chosen PQC is the R_y gate. Thus,

$$\mathcal{F}(\theta) = R_y(\theta\pi/2) \tag{4}$$

The measurement occurs in the Z-basis, given by the expectation of the Pauli matrix, projecting the quantum state into binary outcomes indicating confidence of two classes. The design of the quantum circuit is shown in Fig. 4. The choice of a single-qubit shallow VQC is governed by several key advantages over multi-qubit models in the context of tumor classification. Firstly, this results in a reduction of quantum resource requirements — a feat in itself, as quantum devices are constrained by qubit count and coherence times [106]. Secondly, lower decoherence susceptibility and faster execution is experienced by the circuit, as it minimizes entanglement complexity, enabling better scalability and easier integration with classical pre-processing pipelines [34,65]. Additionally, the single-qubit circuit is more interpretable and demands less quantum gate depth, leading to lower inference latency and reduced cumulative quantum noise compared to deeper multi-qubit alternatives [107]. However, if required, this single-qubit framework may serve as a lightweight and interpretable foundation that is naturally extensible to multi-qubit architectures. The encoding strategy used in the single-qubit model can be generalized to multiple qubits by distributing different image features or statistical descriptors across qubit registers, thus enabling parallel quantum feature maps [43]. Parameterized single-qubit gates and controlled two-qubit operations (e.g., CNOTs) can be stacked into a layered ansatz, increasing expressivity while preserving the original architecture's training logic [108]. This would support scalable classification tasks involving complex, high-dimensional data.

3.5. Training the hybrid network

The training of the hybrid model involves gradient calculation and optimization as in classical models. However, a deterrent here is the inability of quantum devices to store and manipulate the results of



Fig. 4. Quantum circuit used as the classifier in the ResQ model.

intermediate operations, something that is typical of gradient calculation using the method of backpropagation. Thus, the parameter-shift rule [62,81,82] is reverted to, in order to evaluate quantum gradients, where the gradient is calculated using finite differences between shifted values of the parameters. This goes well with the 'black-box' idea of the quantum layer, alleviating the need to know its internal working details.

For calculating errors, partial derivatives of the loss function need to be evaluated in the classical setup. The measured output of the VQC, as given by the expectation in the z-basis, can be expressed as a quantum function $f(\theta)$. The partial derivative of this function, many times, may be expressed as a linear combination of other quantum functions. The parameter-shift rule is based on defining this derivative. The biggest advantage of this rule is that the same quantum circuit can be reused for computing the quantum function as well as its gradient. Mathematically, the parameter-shift rule for gradient evaluation is given as

$$\frac{d}{d\theta} f(\theta) = r\left[f\left(\theta + \frac{\pi}{4r}\right) - f\left(\theta - \frac{\pi}{4r}\right)\right], \tag{5}$$

where, $r = \frac{1}{2}$ for a single qubit Pauli rotation gate. Details of the mathematical foundation of this rule can be found in the Appendix. This gradient-based optimization generalizes naturally to more qubits [62]. The gradient calculation is followed by updating the parameter θ using the gradient descent rule for updating gradients as

$$\theta_{new} = \theta_{old} - \eta\frac{\partial L}{\partial \theta} = \theta_{old} - \eta\frac{\partial L}{\partial f}\frac{\partial f}{\partial \theta}, \tag{6}$$

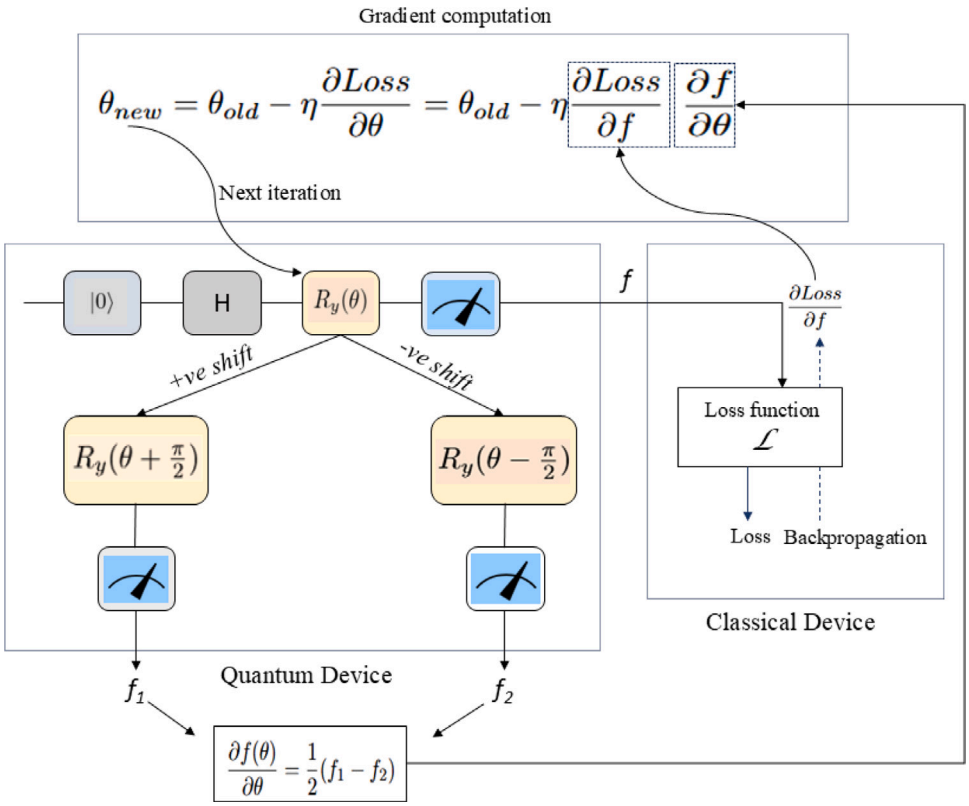


Fig. 5. Gradient calculation using the parameter-shift rule.

where, η is the learning rate and L denotes the loss calculated through the loss function. The optimization continues over several epochs until convergence, minimizing the loss function and achieving accurate classification on unseen data. The training process is summarized in Algorithm 1, and the gradient calculation using the parameter-shift rule has been elaborated in Fig. 5.

By combining the strengths of classical CNNs and quantum computing, the proposed hybrid model, ResQ, promises efficient and accurate classification of histopathological images for breast cancer diagnosis. In addition, this training pipeline, as described above, remains unchanged even when scaling up to more qubits, which might be a requirement on more complex or imbalanced datasets.

4. Experiments

4.1. Dataset description

This study uses two breast histopathology image datasets, focusing on binary classification of malignant and benign cases.

4.1.1. BreakHis dataset

BreakHis [38] is composed of 7909 microscopic images of breast tumor tissue with a resolution of 700 × 460 collected from 82 patients. The images are categorized by magnification levels: 40× (625 benign, 1370 malignant), 100× (644 benign, 1437 malignant), 200× (623 benign, 1390 malignant), and 400× (588 benign, 1232 malignant). Each magnification level includes four types of benign tumors—Adenosis (A), Fibroadenoma (F), Phyllodes Tumor (PT), and Tubular Adenoma (TA)—and four types of malignant tumors—Ductal Carcinoma (DC), Lobular Carcinoma (LC), Mucinous Carcinoma (MC), and Papillary Carcinoma (PC). For this study, the dataset was simplified into binary classes, combining all benign and malignant categories. Sample images of both classes are shown in Fig. 6.

4.1.2. Bioimaging 2015 (BI) dataset

The second dataset, Bioimaging 2015,[2] comprises hematoxylin and eosin (H&E) stain-based images captured at a magnification of 200×, with a pixel scale of 0.42 μm×0.42 μm and a resolution of 2048 × 1536. The dataset includes 251 training images and 36 test images divided into four classes: Normal, Benign, In-Situ Carcinoma, and Invasive Carcinoma. For binary classification, the Normal and Benign classes were merged into a "Non-Cancerous" group, while the In-Situ and Invasive Carcinoma classes formed the "Cancerous" group. Representative samples of these classes are displayed in Fig. 7.

4.2. Experimental setup

The proposed hybrid classical-quantum framework is implemented using Python, with the PyTorch and Qiskit libraries. All experiments are conducted on a system equipped with a 2.60 GHz Intel Xeon E5-2660v3 processor, 128 GB DDR4 ECC RAM, and a 16 GB NVIDIA P5000 GPU. The Aer quantum simulator is employed to execute the VQCs. For a shallow VQC, quantum machine noise/error rates have little or no interference in the model, implying that quantum simulators do a good job of simulating a quantum device [58]. However, for the sake of exhaustive experimentation, real quantum hardware is also resorted to as well for performance evaluation.

For the BreakHis data set, the data are divided into training, validation, and test sets in a 3 : 1 : 1 ratio. Predefined training and test sets are used for the BI dataset. Hyperparameters are duly tuned in order to improve the performance of the considered models. Data augmentation is carried out by transforming the original images through random horizontal and vertical flips, followed by normalization using mean and standard deviation values derived from the pre-trained ImageNet dataset. The learning rate is initialized with 10^{-3} and decays by a factor

[2] http://www.bioimaging2015.ineb.up.pt/dataset.html.

Algorithm 1 Training Procedure of the ResQ model

Input:
T = Training data;
V = Validation data;
E = Total number of Epochs;
B = Batch;
$\mathcal{L}$ = Loss function;
O = Optimizer;

Initialize the pre-trained ResNet-L ($L = 18, 34, 50, 101, 152$) model tweaking the last fully connected layer to produce the Feature $\mathcal{F}$.
for $e = 1$ to **E** **do**
  for $b = 1$ to [**T**/**B**] **do**
    for each image I in batch b **do**
      Extract the feature $\mathcal{F}$ from the pre-trained ResNet-L for image I.
      In the Variational Quantum Classifier (VQC), apply Hadamard Gate (**H**) on a qubit at state $|0\rangle$.
      Take the feature $\mathcal{F}$ as parameter θ to the 1−qubit Pauli rotation gate (R_y) and apply on the superposed outcome of **H**.
      Measure the circuit output in the Pauli-Z basis.
      Calculate the probabilities for each of the two classes.
      Evaluate $\mathcal{L}$ over the circuit output with respect to the label value.
      Perform gradient calculation using parameter-shift rule as per Eq. (5) and Fig. 5.
      Optimize the parameters through the optimizer **O**.
  Evaluate model performance on **V**.
  Calculate validation loss.
  Save model for the minimum value of validation loss.

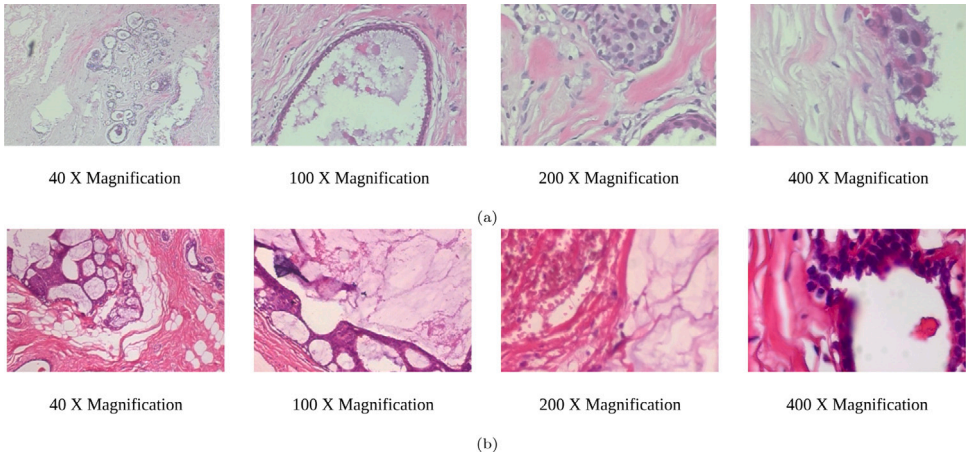


Fig. 6. Sample images from the BreakHis dataset at different magnifications (a) Benign (b) Malignant.

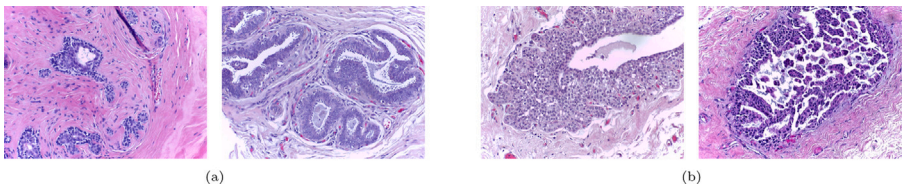


Fig. 7. Sample images from the BI dataset (a) Benign (b) Malignant.

of 0.1 every 7 epochs. The batch size, i,e., the number of samples for each training and validation step in case of ResQ-18, ResQ-34, and ResQ-50, is kept at 64 for training and validation of the BreakHis images. However, due to model complexity and memory limitations, the batch size for the same purpose has to be reduced to 32 for ResQ-101 and ResQ-152. The classical models, though, are executed with a uniform batch size of 64. The selected batch size across architectures as well as cross-validations is 32 for the BI dataset. The validation set

is used to save the model in epochs where the validation loss reaches its minimum. The model is trained for 500 epochs with cross-entropy as the loss function and Stochastic Gradient Descent (SGD) as the optimizer for both datasets. In the quantum portion of the network, the parameter initialization in the PQC occurs with the features extracted by the ResNet architecture. The quantum state in the VQC is initialized with $|0\rangle$. The parameter-shift rule is used for gradient calculation in the quantum network.

4.3. Performance metrics

The model's performance has been evaluated using the following metrics.

1. Accuracy: Accuracy quantifies the proportion of correctly predicted samples out of the total number of predictions and is defined as
$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}},$$
where, TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively.
2. Precision: Precision measures the ratio of true positive predictions to the total predictions made for the positive class, as
$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}.$$
3. Recall (Sensitivity): Recall reflects the model's ability to identify all true positive samples and is defined as
$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}.$$
4. F1-score: F1-score provides a harmonic mean of precision and recall, offering a balance between the two as
$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}.$$
5. Matthew's Correlation Coefficient (MCC): MCC evaluates the correlation between the predicted and actual classes, ranging from −1 (complete disagreement) to +1 (perfect agreement) as
$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}}.$$

In addition to the above-mentioned metrics, statistical analyses are incorporated to evaluate the classifiers — versions of ResNet and ResQ in order to offer better cognizance of the experimental results. For this, the Friedman test [109], a classic rank-based method, has been used as the non-parametric test of choice. For each set of data, ten classifiers (five each for classical and hybrid classical-quantum) and four metrics of evaluation are considered. Also, once the Friedman test yields satisfactory results, a post hoc test, viz., the Nemenyi test [110], is carried out to compare the classifiers pair-wise.

4.4. Comparison with other hybrid quantum neural networks

A notable work in the domain of transfer-learned classical-quantum hybrid neural networks is the work by Mari et al. in 2020 [56]. Deemed as a pioneering proposal in the context of augmenting a pre-trained classical neural network with a variational quantum classifier at the final layer, this work serves as the baseline for many a subsequent applications such as classifying faces wearing face-mask [87], detection of breast cancer [58,73], diagnosis of diabetic retinopathy [86,111], classifying presence of diseases in maize leaf [89], to name a few. The Classical-to-Quantum (CQ) variation of the aforementioned model has been implemented on the datasets of concern in this work to compare its performance with the proposed approach. Along with accuracy, precision, recall, F1-score, and MCC, an endeavor to develop insight into the computational efficiency is also made by measuring the time required to execute the respective networks on the same device with similar execution parameters.

4.5. Circuit complexity metrics

The complexity of the quantum circuit is evaluated using the following measures.

1. Circuit Depth: An intuitive measure is the circuit depth that counts the time steps required for the execution of a quantum circuit [112] under the assumption of parallel gate operations. Lower depth is desirable for near-term intermediate-scale quantum devices.
2. Gate Count: The overall size and complexity of the circuit are reflected by this measure, which tallies the total number of gates of a circuit.
3. Algebraic Complexity: This measure calculates the minimum number of non-trivial single-qubit or two-qubit gates required to implement the circuit's unitary operations [113].

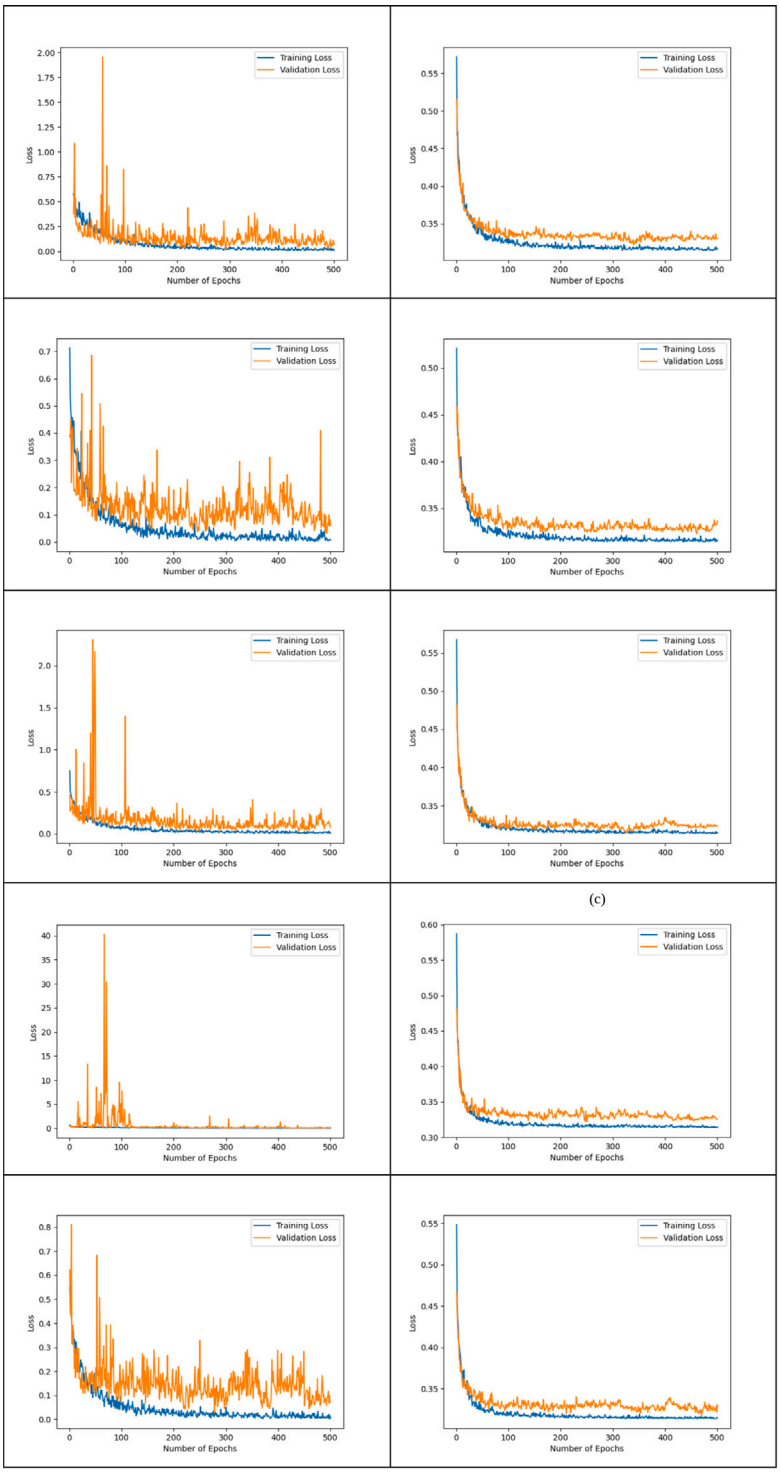
These metrics are crucial for assessing the practicality of quantum circuits on NISQ devices.

5. Results and discussion

In this section, the results of experiments conducted on the BreakHis and BI datasets are presented. The training and validation loss curves for both the ResQ-L ($L = 18, 34, 50, 101, 152$) and its classical counterparts on the BreakHis dataset at 400× magnification are shown in Fig. 8. Executed for 500 epochs each, both sets of curves exhibit fair convergence, with no evident overfitting or underfitting, indicating stable learning across the models. However, compared to the classical network, the validation loss for the hybrid network, follows the training loss curves more closely, demonstrating its better learning behavior.

5.1. Performance comparison of classical neural network and hybrid classical-quantum network

Table 2 and Fig. 9 present a comparative analysis of the performance metrics for the classical ResNet-L and ResQ-L models on the BreakHis dataset. Every ResNet-L and ResQ-L model has been executed 5 times on each dataset, and the mean and standard deviations of the obtained percentage metric values have been showcased. Each of the hybrid models, ResQ-50, ResQ-101, and ResQ-152, displays the highest accuracy percentages of 99.83% for the 40× magnified BreakHis images and 99.68% for the 100× magnified images. ResQ-18 also shows a slight improvement in accuracy over its classical counterpart for both magnifications. In case of the 200× magnified images, 99.59% is the highest accuracy displayed by the ResQ models 34 and 50. Lastly, ResQ-50 gives an accuracy of 98.49% for 400× magnification — the highest among all the ResNet-L and ResQ-L models for the concerned data. Though the accuracy percentages are significantly higher in the case of the hybrid models than the classical ones, barring a case or two (ResNet-34 for 100× images, ResNet-152 for 200× images), the more important metric when it comes to cancer diagnosis is the recall, also known as sensitivity. It denotes the percentage figure as to how many actual cancerous cases (true positives) the model reports correctly. This is so because a false negative, that is, identifying a malignant lesion as non-cancerous, would result in progression of the disease without medical treatment, leading to fatal consequences. Thus, ensuring a higher recall value is of utmost importance. ResQ-50 has been found to yield the best recall percentages for three out of the four magnifications – 40×, 200×, 400× – of breast histopathology images. For the remaining magnification configuration, i.e., 100×, ResQ-152 gives the best recall, thereby establishing the superiority of each of the ResQ-L architectures in cancer detection. The downside of emphasizing only a high recall is classifying too many healthy patients as sick (false positives). Precision takes care of such cases, ensuring accuracy of the reported positive predictions, reducing the risk of false alarms. ResQ-50 again champions



(a) Classical Loss Curves (b) Hybrid Loss Curves

Fig. 8. Training and validation loss curves on the BreakHis dataset at 400× magnification level.

the percentage figures of precision for the magnification factors 100× and 400×. For 40× and 20×, ResQ-101 and ResQ-34, respectively, do the job. Striking a balance between the two — precision and recall, is the F1-score, a metric that combines both precision and recall, thereby providing a balance between neither missing out cancer identification nor over-diagnosing. Across magnifications, the ResQ-L models exhibit higher F1-scores than the ResNet-L ones. In particular, best figures are showcased by ResQ-50 for all four magnification factors. The above discussion clearly indicates that the combination of the transfer-learned ResNet-50 with the VQC, viz, ResQ-50 proves to be the most proficient in classifying BreakHis images effectively.

To further validate this finding, the same experiments have been conducted on the BI dataset employing 5-fold cross validation. The results indicate that ResQ-34 and ResQ-50 respectively achieve almost 3% and 4% improvements in accuracy, recall, and F1-score figures over ResNet-34 and ResNet-50. Though ResNet-101 and ResQ-101 display similar results on this dataset, ResQ-152 again surpasses ResNet-152, yielding the best figures among the classifiers across all quantitative

Table 2
Comparison of accuracy, precision, recall, F1-score (expressed in percentages), and MCC using classical and hybrid classical-quantum networks on the BreakHis dataset at different magnifications. Results of the best-performing models have been marked in bold.

Dataset	Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	MCC
BreakHis 40×	ResNet-18	99.33 ± 0.63	99.03 ± 0.89	99.44 ± 0.57	99.23 ± 0.73	0.98
	ResQ-18	99.58 ± 0.14	99.41 ± 0.19	99.62 ± 0.20	99.51 ± 0.17	0.99
	ResNet-34	99.58 ± 0.52	99.412 ± 0.70	99.624 ± 0.50	99.52 ± 0.60	0.99
	ResQ-34	99.58 ± 0.52	99.41 ± 0.70	99.62 ± 0.50	99.52 ± 0.60	0.99
	ResNet-50	99.75 ± 0.25	99.67 ± 0.30	99.75 ± 0.30	99.71 ± 0.29	0.99
	ResQ-50	**99.83 ± 0.29**	99.74 ± 0.45	**99.88 ± 0.21**	**99.81 ± 0.33**	**0.99**
	ResNet-101	99.58 ± 0.38	99.42 ± 0.59	99.62 ± 0.33	99.52 ± 0.44	0.98
	ResQ-101	**99.83 ± 0.14**	**99.81 ± 0.20**	99.81 ± 0.20	**99.81 ± 0.17**	**0.99**
	ResNet-152	98.83 ± 1.81	98.58 ± 2.13	98.71 ± 2.07	98.64 ± 2.10	0.97
	ResQ-152	**99.83 ± 0.14**	99.74 ± 0.23	**99.88 ± 0.10**	**99.81 ± 0.17**	**0.99**
BreakHis 100×	ResNet-18	98.88 ± 0.37	98.45 ± 0.46	98.98 ± 0.43	98.71 ± 0.42	0.97
	ResQ-18	99.44 ± 0.37	99.18 ± 0.47	99.52 ± 0.39	99.35 ± 0.42	0.98
	ResNet-34	99.6 ± 0.14	99.37 ± 0.22	99.68 ± 0.10	99.53 ± 0.16	0.99
	ResQ-34	99.2 ± 0.50	99.29 ± 0.49	98.86 ± 0.77	99.06 ± 0.59	0.98
	ResNet-50	99.6 ± 0.37	99.5 ± 0.47	99.57 ± 0.38	99.53 ± 0.43	0.99
	ResQ-50	**99.68 ± 0.14**	**99.7 ± 0.22**	99.56 ± 0.10	**99.63 ± 0.16**	**0.99**
	ResNet-101	99.6 ± 0.14	99.5 ± 0.10	99.57 ± 0.22	99.53 ± 0.16	0.99
	ResQ-101	**99.68 ± 0.14**	99.63 ± 0.19	99.63 ± 0.19	**99.63 ± 0.16**	**0.99**
	ResNet-152	99.6 ± 0.14	99.57 ± 0.30	99.5 ± 0.24	99.54 ± 0.16	0.99
	ResQ-152	**99.68 ± 0.14**	99.56 ± 0.10	**99.7 ± 0.22**	**99.63 ± 0.16**	**0.99**
BreakHis 200×	ResNet-18	99.01 ± 0.25	98.99 ± 0.42	98.7 ± 0.22	98.84 ± 0.29	0.98
	ResQ-18	99.34 ± 0.28	99.45 ± 0.33	99.01 ± 0.34	99.23 ± 0.33	0.98
	ResNet-34	99.34 ± 0.38	99.38 ± 0.49	99.09 ± 0.45	99.23 ± 0.44	0.99
	ResQ-34	**99.59 ± 0.28**	**99.63 ± 0.33**	99.41 ± 0.33	**99.52 ± 0.33**	**0.99**
	ResNet-50	99.34 ± 0.14	99.3 ± 0.31	99.16 ± 0.11	99.23 ± 0.17	0.98
	ResQ-50	**99.59 ± 0.14**	99.48 ± 0.10	**99.55 ± 0.23**	**99.52 ± 0.17**	**0.99**
	ResNet-101	99.34 ± 0.14	99.37 ± 0.23	99.09 ± 0.10	99.23 ± 0.17	0.98
	ResQ-101	99.42 ± 0.14	99.44 ± 0.21	99.22 ± 0.42	99.33 ± 0.17	0.99
	ResNet-152	99.5 ± 0.43	99.5 ± 0.56	99.35 ± 0.43	99.42 ± 0.50	0.99
	ResQ-152	99.34 ± 0.38	99.17 ± 0.60	99.3 ± 0.28	99.23 ± 0.44	0.98
BreakHis 400×	ResNet-18	96.03 ± 1.31	95.56 ± 1.78	94.98 ± 2.05	95.24 ± 1.85	0.90
	ResQ-18	97.27 ± 0.43	96.53 ± 0.84	97.37 ± 0.13	96.92 ± 0.45	0.94
	ResNet-34	96.80 ± 2.14	96.16 ± 2.29	96.57 ± 2.62	96.36 ± 2.45	0.93
	ResQ-34	98.12 ± 0.17	97.68 ± 0.39	98.08 ± 0.59	97.86 ± 0.19	0.95
	ResNet-50	97.36 ± 1.07	97.06 ± 1.07	96.92 ± 1.39	96.99 ± 1.23	0.94
	ResQ-50	**98.49 ± 0.65**	**98.15 ± 0.63**	**98.43 ± 0.88**	**98.29 ± 0.75**	**0.96**
	ResNet-101	97.45 ± 1.23	96.95 ± 1.58	97.29 ± 1.17	97.12 ± 1.38	0.94
	ResQ-101	97.93 ± 1.07	97.39 ± 1.33	97.94 ± 1.10	97.65 ± 1.21	0.95
	ResNet-152	96.14 ± 1.07	95.09 ± 1.06	96.31 ± 1.44	95.65 ± 1.22	0.92
	ResQ-152	97.65 ± 1.17	96.98 ± 1.41	97.73 ± 1.21	97.34 ± 1.32	0.95

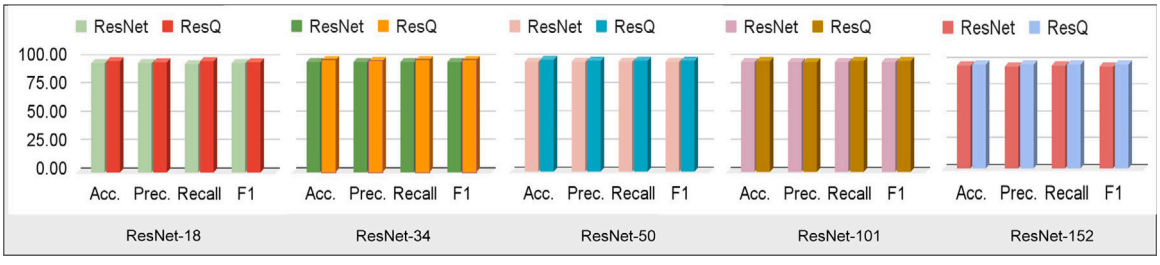


Fig. 9. Comparison of accuracy, precision, recall, F1-score (expressed in percentages and abbreviated as Acc., Prec., F1) of ResQ-L with its classical counterpart ResNet-L models on 400× magnified BreakHis dataset. Here L represents different variants of ResNet, i.e. 18, 34, 50, 101, 152.

metrics. The highest MCC value too is showcased by ResQ-152 at 0.79. These results are summarized in Table 3 and visually represented in Fig. 10, reinforcing the superiority of the hybrid approach over the classical one.

The Friedman test is applied on the performance metrics, accuracy, precision, recall, and F1-score results, to compare the individual classifiers. The classifiers are tested for being similar under the null hypothesis. Likewise, the alternative hypothesis states that the classifiers bear significant dissimilarities between themselves. The null hypothesis stands rejected when the value of the test statistic, *chi-square* in the case of the Friedman test, exceeds the critical value for the chosen level of significance α. The level of significance α is set to 0.05. For BreakHis 40×, 100×, 200×, 400× and BI data, the *chi-square* values are respectively obtained as **38.25**, **20.67**, **30.44**, **82.16**, and **30.03**. The critical value of *chi-square* for $\alpha = 0.05$ and $(k-1)$ degrees of freedom, where k is the number of classifiers (=10, here), is **16.919**. Evidently, the null hypothesis is rejected in all cases, indicating that there is significant differences amid classifiers. This leads to the next non-parametric post hoc test, Nemenyi

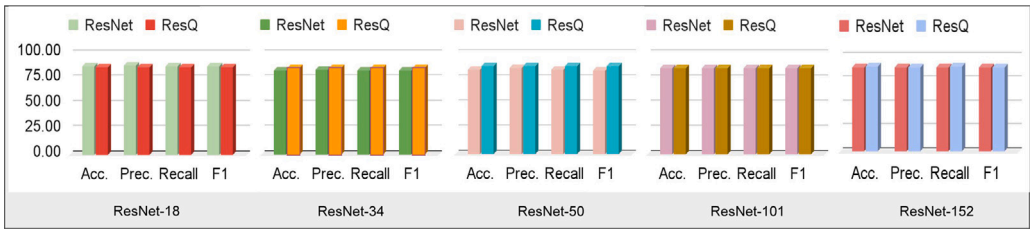


Fig. 10. Comparison of accuracy, precision, recall, F1-score (expressed in percentages and abbreviated as Acc., Prec., F1) of ResQ-L with its classical counterpart ResNet-L models on the BI dataset. L denotes the backbone ResNet configurations, viz, 18, 34, 50, 101, 152.

Table 3
Comparison of accuracy, precision, recall, F1-score (expressed in percentages), and MCC using classical and hybrid classical-quantum networks on the BI dataset using 5-fold cross validation. Results of the best-performing models have been marked in bold.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	MCC
ResNet-18	87.04 ± 3.21	87.78 ± 2.36	87.04 ± 3.21	86.95 ± 3.36	0.76
ResQ-18	86.11 ± 0	86.22 ± 0	86.11 ± 0	86.10 ± 0	0.71
ResNet-34	82.407 ± 4.24	83.371 ± 3.98	82.407 ± 4.24	82.265 ± 4.35	0.66
ResQ-34	85.18 ± 1.60	85.4 ± 1.43	85.18 ± 1.60	85.16 ± 1.63	0.70
ResNet-50	83.33 ± 4.24	85.06 ± 3.99	83.33 ± 4.24	83.13 ± 4.41	0.67
ResQ-50	87.04 ± 1.60	87.42 ± 1.35	87.04 ± 1.60	87 ± 1.64	0.74
ResNet-101	85.18 ± 1.61	85.4 ± 1.43	85.18 ± 1.61	85.16 ± 1.63	0.71
ResQ-101	85.18 ± 3.21	85.32 ± 3.10	85.18 ± 3.21	85.17 ± 3.22	0.70
ResNet-152	87.96 ± 1.6	88.16 ± 1.7	87.96 ± 1.6	87.95 ± 1.6	0.77
ResQ-152	**88.96 ± 2.71**	**88.58 ± 3.38**	**88.96 ± 2.71**	**87.99 ± 2.36**	**0.79**

test, that is engaged to determine where the differences that are of significance, exist.

The Nemenyi test evaluates the average rankings of classifiers to determine if there are statistically meaningful differences between pairs. It calculates the differences in mean ranks for each pair and compares these to a Critical Distance (CD), which is obtained from the Nemenyi distribution table based on a chosen significance level α. In this case, the critical distance is **6.77**, for 9 degrees of freedom with α at 0.05. In the analysis of the BreakHis dataset at 40× magnification, pairwise comparisons indicate that ResQ-101 and ResQ-152 significantly outperform ResNet-18 and ResNet-152. At 100× magnification, no pairwise differences surpass the critical threshold, suggesting no statistically significant distinctions. For images at 200× magnification, ResQ-152 performs significantly better than ResNet-18 and ResNet-34. At 400×, both ResQ-34 and ResQ-50 significantly outperform ResNet-18, ResNet-34, and ResNet-152. On the BI dataset, ResQ-152 demonstrates a notable performance edge over ResNet-50.

Overall, these results consistently highlight the superior performance of certain ResQ models compared to traditional ResNet classifiers across different datasets and magnification levels.

Further experiments have been conducted on diverse data splits on all magnification factors of the BreakHis dataset. Instead of the random 3 : 1 : 1 train, validation, and test split, a patient-wise split, ensuring that no patients are common among the train, validation, and test sets, is done, followed by evaluating this differently split data using ResQ-50 — the best model established and its corresponding classical counterpart. Table 4 reports the results obtained. The percentage figures coupled with the MCC, as expected, display a drop from the previous random 3 : 1 : 1 data split highlighting the more realistic generalization performance of the model under this stringent setup. Another study included distributing patient records uniformly across the three sets of data used for training, testing, and validation purposes, so that every set is made to contain data collected from a common patient. The results thus obtained, on evaluation of this uniform patient-data distribution employing ResNet-50 and ResQ-50, have been showcased in Table 5. The uniform vs. original random split display negligible variations ranging from 0.00 to 0.55% in ResQ-50 average accuracy figures and from 0.11 to 0.91% for the same in the case of ResNet-50, indicating model stability.

Table 6 presents some sample images from the BreakHis dataset (400× magnification) as well as the BI dataset that demonstrate misclassifications and correct classification results by the individual models. There are thirteen images for which ResQ-18 provides correct classification, but ResNet-18 provides wrong classification outcomes. However, the reverse is true for only eight images. There are only four instances where ResQ-34 has misclassified those very images which had been correctly classified by ResNet-34, whereas for thirty-five images, ResQ-34 has identified the correct class, but ResNet-34 has failed. Not a single such instance exists where ResNet-50 has correctly classified an image which ResQ-50 has failed to classify. Similarly, on the BI dataset too, though two images wrongly classified by ResNet-50 are correctly classified by ResQ-50, no such image exists that ResNet-50 correctly classifies, but ResQ-50 does not. In summary, for both the BreakHis and BI datasets, the hybrid classical-quantum network consistently outperforms the classical ResNet models in accuracy, precision, recall, F1-score, and MCC, with improvements ranging from 2% to 4%, depending on the dataset and magnification factor — a fact substantiated by statistical tests as well.

5.2. Performance comparison with recent state-of-the-art models for BreakHis

To demonstrate the superiority of the proposed method, it is compared with several benchmark models used for breast cancer detection on the BreakHis dataset. Among the most recent approaches, Addo et al. [114] introduced a hybrid lightweight AI framework called BCHI-CovNet, which incorporates a multi-head self-attention mechanism along with multi-scale depth-wise separable convolution and pooling modules. This innovative framework efficiently captures both low and high-resolution features, yielding impressive accuracies: 99.15% for 40×, 99.08% for 100×, 99.22% for 200×, and 98.87% for 400× magnification on the BreakHis dataset. Kolla et al. [115] proposed an integrated approach using the Swin Transformer V2 architecture, designed to classify both Magnification-Dependent (MD) and Magnification-Independent (MI) images. The MD BreakHis results achieved accuracy percentages of 99.73% at 40×, 99.55% at 100×, 98.45% at 200×, and 98.88% at 400×. Additionally, Chattopadhyay et al. [116] introduced the Multi-scale Dual Residual Recurrent Network (MTRRE-Net), an

Table 4
Accuracy, precision, recall, F1-score (expressed in percentages), and MCC measured for ResNet-50 and ResQ-50 after splitting the BreakHis dataset patient-wise, i.e., train, test, and validation sets do not have any patient data in common.

Dataset	Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	MCC
BreakHis 40×	ResNet-50	95.19 ± 3.35	95.38 ± 1.99	93.78 ± 5.86	94.26 ± 4.27	0.93
	ResQ-50	94.98 ± 1.14	93.35 ± 0.99	96.03 ± 0.76	94.43 ± 0.94	0.94
BreakHis 100×	ResNet-50	91.99 ± 1.92	90.16 ± 2.12	94.72 ± 1.56	91.87 ± 2.09	0.83
	ResQ-50	93.27 ± 1.90	89.74 ± 2.19	94.66 ± 1.14	91.47 ± 2.03	0.92
BreakHis 200×	ResNet-50	91.68 ± 3.76	90.18 ± 4.14	93.05 ± 2.55	91.04 ± 3.86	0.80
	ResQ-50	89.67 ± 0.64	87.81 ± 0.58	91.34 ± 0.32	88.90 ± 0.61	0.79
BreakHis 400×	ResNet-50	84.75 ± 1.86	83.71 ± 2.73	83.15 ± 0.69	83.28 ± 1.62	0.67
	ResQ-50	84.75 ± 0.58	82.59 ± 0.75	85.01 ± 1.54	83.4 ± 0.84	0.70

Table 5
Comparison of accuracy, precision, recall, F1-score (expressed in percentages), and MCC measured after uniformly distributing patient records across train, validation, and test sets of the BreakHis data for all magnifications.

Dataset	Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	MCC
BreakHis 40×	ResNet-50	99.01 ± 0.31	99.4 ± 0.36	98.93 ± 0.36	99.16 ± 0.36	0.98
	ResQ-50	99.28 ± 0.16	99.21 ± 0.22	98.48 ± 0.24	98.84 ± 0.19	0.98
BreakHis 100×	ResNet-50	99.82 ± 0.15	99.8 ± 0.21	99.8 ± 0.21	99.8 ± 0.17	0.99
	ResQ-50	99.91 ± 0.15	99.86 ± 0.24	99.94 ± 0.11	99.9 ± 0.17	0.99
BreakHis 200×	ResNet-50	99.45 ± 0.27	99.45 ± 0.32	99.29 ± 0.32	99.37 ± 0.31	0.99
	ResQ-50	99.59 ± 0.35	99.65 ± 0.22	99.14 ± 0.62	99.52 ± 0.42	0.99
BreakHis 400×	ResNet-50	98.27 ± 0.88	98.15 ± 0.80	97.91 ± 1.21	98.03 ± 1.01	0.96
	ResQ-50	98.47 ± 0.48	98.16 ± 0.64	98.39 ± 0.43	98.27 ± 0.54	0.98

end-to-end deep learning model. This approach achieved accuracy percentages of 97.12%, 95.22%, 96.85%, and 97.81% for the respective magnifications of 40×, 100×, 200×, and 400× on the BreakHis dataset. The performance results for these models are summarized in Table 7. From the comparison, it is evident that the proposed method performs comparably to the current state-of-the-art models, for the BreakHis dataset, demonstrating its competitive efficacy in breast cancer detection. For the BI data too, ResQ-152 achieves comparable performance with the work reported in [5], wherein, unlike our work, cross validation or standard deviations have not been considered.

5.3. Comparison with other hybrid classical-quantum networks

In this section, the performance of the proposed hybrid classical-quantum network has been compared with another hybrid classical-quantum network, the one introduced by Mari et al. [56], abbreviated henceforth as AM. The performance metrics of interest — accuracy, precision, recall, F1-score, and MCC — for both ResQ and AM are presented in Table 8 for the BreakHis dataset at 400× magnification as well as for the BI dataset. Additionally, the execution time required for training the respective networks has been highlighted, as execution time is a crucial factor when working with hybrid classical-quantum models. Given that quantum resources are still limited and classical hardware is often used to simulate quantum phenomena, the execution time tends to be relatively high. Therefore, it is important to consider the time taken for training these hybrid networks before drawing conclusions about their effectiveness in the classification tasks. The proposed ResQ network shows an improvement of at least 10% in accuracy compared to the network AM on the BreakHis dataset. Notably, the ResQ model also requires less training time. The corresponding confusion matrices, shown in Fig. 11, reveal the true and predicted results with respect to positive and negative diagnosis outcomes. The principal diagonal values respectively denote true positive and true negative predictions – larger values indicating model efficiency. On the contrary, the off-diagonal figures reveal misclassifications – false negatives (malignant lesions falsely classified as negative) in the first row, second column, and false positives (non-cancerous patients misclassified as positive) in the second row, first column. As indicated in Fig. 11, a significant portion of data falls under the false positive and false negative categories for the network AM. In contrast, the confusion matrices in Fig. 12 for the proposed model show minimal misclassifications.

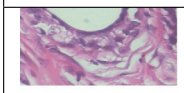
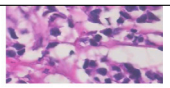
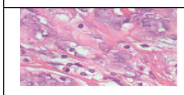
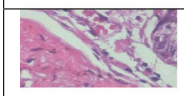
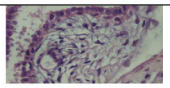
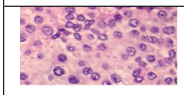
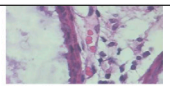
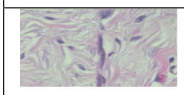
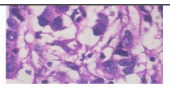
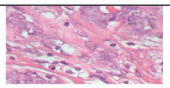
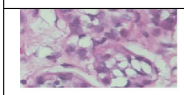
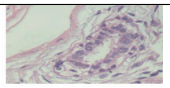
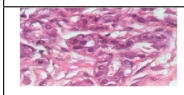
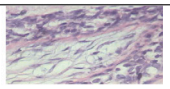
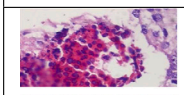
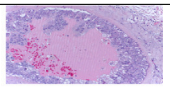
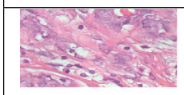
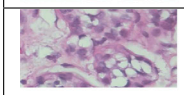
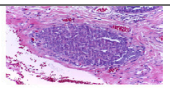
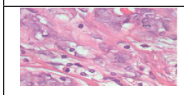
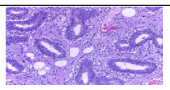
For the BI dataset, the network AM fails to perform adequately, as evidenced by lower accuracy percentages. On the other hand, the proposed hybrid network efficiently classifies the BI dataset, achieving better results in nearly half the execution time across different versions of ResNet.

The two networks, along with other similar works [55,58,87,123,124] based on the variational circuit analysis metrics described in Section 4.5, are duly compared. The VQC in the network AM has a circuit depth of 6, whereas the VQC in the proposed ResQ network has a circuit depth of just 1, offering the advantage of a shallower circuit that requires only a single time step for execution. Additionally, ResQ network's VQC operates with a single qubit, while the network AM uses 4 qubits. Since qubits are inherently volatile and susceptible to noise, fewer qubits lead to simplified error correction and reduced susceptibility to errors. The next metric for comparison is the gate count. The VQC in ResQ performs significantly better in this regard, requiring only 3 gates compared to AM's 15 gates. Furthermore, the algebraic complexities of the respective VQCs are assessed. The VQC in the proposed ResQ network involves three non-trivial single-qubit gates: the Hadamard gate, the rotation gate, and the Pauli-Z gate. On the other hand, the VQC in the AM architecture not only includes these three non-trivial single-qubit gates, but also incorporates a two-qubit Controlled-NOT (CNOT) gate, increasing its algebraic complexity. Similarly, COVID-19 detection by Acar et al. [123] from CT images uses a hybrid network with circuit depth of 6, 17 gates, and 4 single-qubit and double-qubit gates. A summary of the comparative hardware requirements for hybrid networks is presented in Table 9.

5.4. Execution on a real quantum processor

Quantum hardware with recourse to IBM quantum processors has also been engaged to perform experiments on the hybrid models. Due to the limited accessibility to real-time quantum processors, the hybrid classical-quantum architecture that held the most promising consequences, i.e., ResQ-50, is tested on the IBM quantum device — *ibm_brisbane*. Out of the different magnifications of the breast histopathology images, 40× is selected as it achieved the best results in terms of performance metrics. On three repetitive runs, slight drop in all the metrics — average accuracy, precision, recall, F-1 score, and MCC, as obtained on the simulator, was observed which may be attributed to

Table 6
Some example images from the BreakHis 400× and BI datasets along with their classification results using the classical and hybrid models. Here, ✓& ✗ respectively indicate proper and wrong classification by the respective model.

Image	Model		Image	Model	
BreakHis	ResNet-18	ResQ-18	BreakHis	ResNet-34	ResQ-34
	✓	✓		✓	✓
	✓	✗		✓	✗
	✗	✓		✗	✓
	✗	✗		✗	✗
Image	Model		Image	Model	
BreakHis	ResNet-50	ResQ-50	BreakHis	ResNet-101	ResQ-101
	✓	✓		✓	✓
NA	✓	✗		✓	✗
	✗	✓		✗	✓
	✗	✗		✗	✗
Image	Model		Image	Model	
BreakHis	ResNet-152	ResQ-152	BI	ResNet-50	ResQ-50
	✓	✓		✓	✓
	✓	✗	NA	✓	✗
	✗	✓		✗	✓
	✗	✗		✗	✗

noise intervention in the quantum hardware owing to the fact that NISQ devices are still not robust against noise. Table 10 showcases the figures.

5.5. Execution time and time complexity analysis

To better understand the computational efficiency of the proposed ResQ model, we analyze both its theoretical time complexity and its actual runtime during training. ResQ combines a pretrained ResNet for classical feature extraction with a streamlined quantum classifier built using a single-qubit VQC. This hybrid structure is carefully designed to ensure a balance between accuracy and speed, especially in the context of NISQ devices.

In terms of theoretical complexity, the classical part of the model, i.e., the ResNet, scales according to the number of layers and operations it performs. We denote this as $\mathcal{O}(C_{\text{resnet}})$, which varies depending on the chosen ResNet depth. The quantum part, however, is exceptionally lightweight. It uses just one qubit and three basic quantum gates – Hadamard, R_y, and Pauli-Z – arranged in a shallow, single-layer circuit. Since there is no entanglement or interaction between multiple qubits, the quantum circuit has a fixed computational cost per input, or $\mathcal{O}(1)$, regardless of the input size.

This is in contrast to many other hybrid quantum models, where the quantum component is much more involved. These models often rely on multiple qubits, greater circuit depth, and a higher number of gates, particularly entangling gates like CNOT. In general, such quantum circuits can be represented with a complexity of $\mathcal{O}(d \cdot g \cdot q^2)$, where d is the circuit depth, g is the number of gates per layer, and q is the number of qubits. The quadratic dependency arises because of interactions between qubits, which increase execution time and

Table 7
Comparison of the state-of-the-art models in terms of accuracy on the BreakHis and BI datasets. Best results are marked in bold.

Comparison of some recent state-of-the-art and proposed method						
Reference, Year	Methodology for classification	Dataset	Accuracy (%) by magnification factor			
			40×	100×	200×	400×
[117], 2018	Incremental Boosting DCNN model fusion	BreakHis	95.1	96.3	96.9	93.8
[118], 2019	FCNN combined with Bi-LSTM		95.69	93.61	96.32	94.29
[119], 2020	VGGEt-16 with SVM and RF		94.11	95.12	97.01	94.96
[116], 2022	MTRRE-Net74		97.12	95.22	96.85	97.81
[120], 2022	DRDA-Net		95.72	94.41	97.43	96.84
[121], 2023	GLNet		90.1	92.32	91.98	91.51
[114], 2024	BCHI-CovNet		99.15	99.08	99.22	98.87
[115], **2024**	Modified Swin-Transformer V2 architecture		99.73	99.55	98.45	**98.88**
Proposed method	**ResNet-50 + VQC**		**99.83**	**99.68**	**99.59**	98.49
[122], 2017	CNN + SVM	BI			83.3	
[5], **2023**	**InceptionNet-V3, DenseNet-201 and ResNet-34 fusion using GC**				**91.67**	
Proposed method	**ResNet-152+ VQC**				88.96	

Table 8
Comparison of the performance metrics calculated using Mari et al. (abbreviated as AM-L) and the proposed ResQ-L on the BreakHis (at 400×magnification images) and BI datasets. Here, L represents different variants of ResNet, i.e., 18, 34, 50, 101, 152. The execution times (in minutes) have been recorded from the machine described in the experimental setup. Results of the best-performing models have been marked in bold.

Comparison metrics	Different configurations of AM and ResQ									
	AM-18	ResQ-18	AM-34	ResQ-34	AM-50	ResQ-50	AM-101	ResQ-101	AM-152	ResQ-152
BreakHis 400×										
Accuracy	85.31	97.27	85.59	98.12	85.88	**98.49**	85.31	97.93	87.29	97.65
Precision	83.65	96.53	85.33	97.68	84.76	**98.15**	84.43	97.39	86.80	96.98
Recall	82.35	97.37	80.98	98.08	82.32	**98.43**	81.23	97.94	83.59	97.73
F1-score	82.94	96.92	82.59	97.86	83.34	**98.29**	82.49	97.65	84.89	97.34
MCC	0.66	0.94	0.66	0.95	0.67	**0.96**	0.66	0.95	0.70	0.95
Execution time (mins)	715	708	667	690	701	**607**	701	661	676	675
	AM-18	ResQ-18	AM-34	ResQ-34	AM-50	ResQ-50	AM-101	ResQ-101	AM-152	ResQ-152
BI										
Accuracy	52.78	86.11	44.44	85.18	58.33	87.04	69.44	85.18	66.67	**87.96**
Precision	53.27	86.22	44.16	85.4	77.27	87.42	69.50	85.32	67.53	**88.58**
Recall	52.78	86.11	44.44	85.18	58.33	87.04	69.44	85.18	66.67	**87.96**
F1-score	50.92	86.10	43.75	85.16	49.58	87	69.42	85.17	66.25	**87.89**
MCC	0.04	0.71	0.14	0.70	0.30	0.74	0.37	0.70	0.32	**0.79**
Execution time (mins)	539	**208**	517	391	405	268	441	223	455	230

Table 9
Comparison of hardware requirements of the hybrid networks.

Circuit analysis metrics	[56]	[123]	[87]	[55]	[58]	[124]	Proposed network
Circuit depth	6	6	10	4	**1**	3	**1**
Gate count	15	17	20	14	19	19	**3**
Algebraic complexity	4	4	4	**3**	4	4	**3**
Number of qubits	4	4	4	4	4	4	**1**

Table 10
Quantitative metrics acquired after executing ResQ-50 on quantum processor over the test set of 40× magnified BreakHis images.

Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	MCC
99	98.63	99.05	98.84	0.98

sensitivity to quantum noise — two major limitations in real-world quantum hardware today.

To evaluate the efficiency of our design in practice, we measure how long each part of the model takes to run during training. This profiling is done using the ResQ-50 variant on the BreakHis dataset at 400× magnification. It is found that the ResNet component consumes approximately 282.24 min, while the VQC layer takes about 190.86 min. The full training session lasted around 607 min. From this breakdown, it is clear that ResNet accounts for nearly half of the total execution time, and the quantum part contributes roughly a third.

For further perspective, we compare these figures with those from the previously discussed well-known hybrid model proposed by Mari et al.- AM, which incorporates a more complex VQC with four qubits, deeper layers, and more gates. The ResNet component in AM-50 consumes approximately 212 min, whereas the VQC takes about 380 min to execute, under the same conditions as ResQ-50. These findings, along with the time required to test a single image on the real quantum processor — *ibm_brisbane* using AM-50 and ResQ-50, have been presented in Table 11. The performance gains of ResQ are achieved without sacrificing accuracy, thanks to its intentionally simple quantum design.

In summary, ResQ not only improves classification performance but also does so in a highly efficient and hardware-friendly manner. The profiling results confirm that most of the computational load lies in the ResNet backbone, but the lightweight quantum layer still contributes meaningfully. Together, the theoretical analysis and empirical results demonstrate that ResQ stands out as a practical and computationally efficient hybrid model suitable for real-world applications.

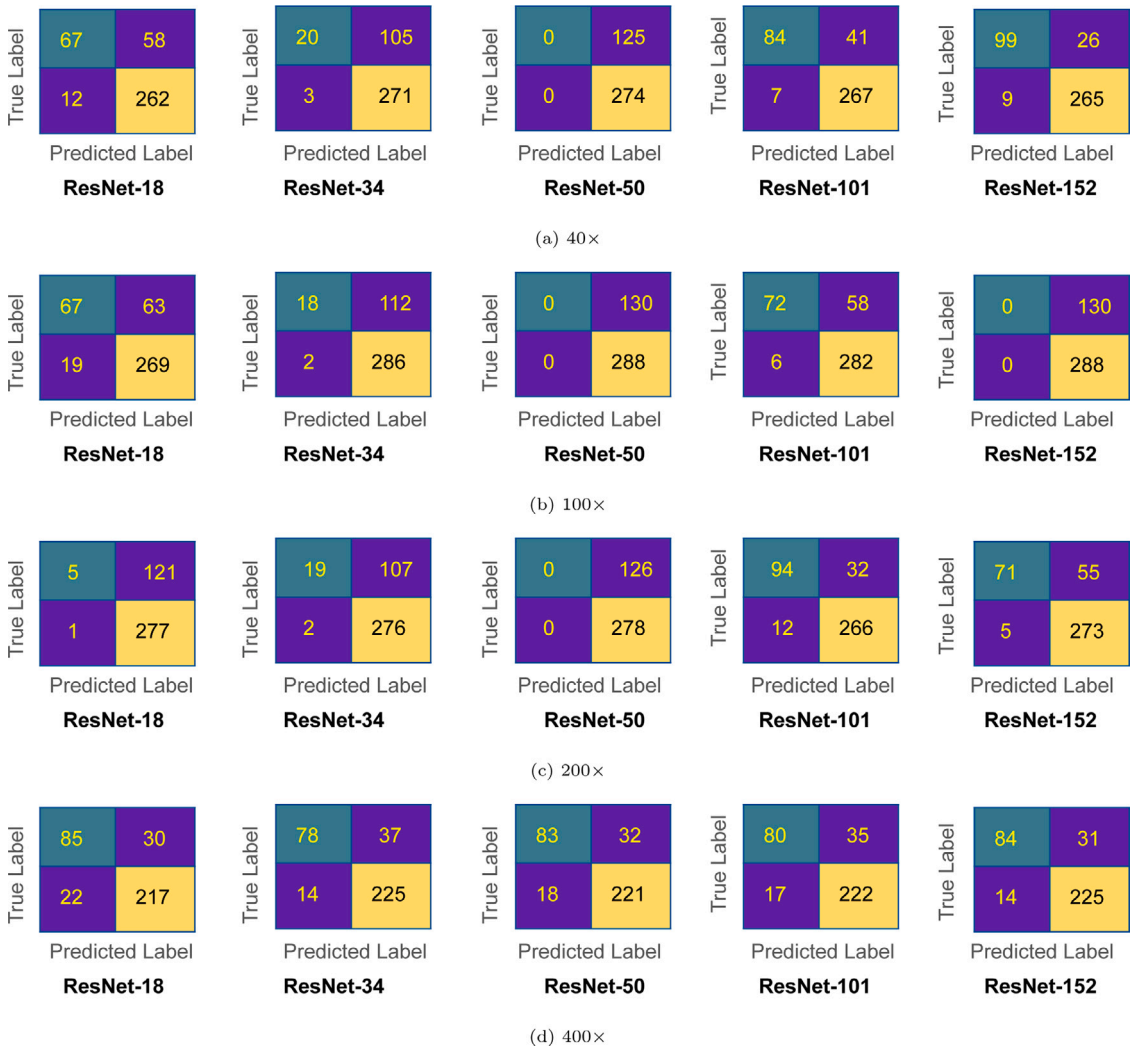


Fig. 11. Confusion matrix for images of the BreakHis test dataset at different magnifications (a) 40× (b) 100× (c) 200× (d) 400× classified using the network architecture (AM) by Mari et al. [56].

Table 11
Comparison of ResQ-50 with AM-50 on the BreakHis 400× dataset in terms of time efficiency.

Methodology	No. of qubits	Time complexity	Component-wise training time on simulator (mins)	Inference time per image on quantum processor (s)
AM-50 [56]	4	$O(d.g.q^2)$	ResNet - 203, VQC - 338.55	14
ResQ-50	1	$O(1)$	ResNet - 282.24, VQC - 190.86	2

6. Conclusion

In this paper, ResQ, a novel hybrid classical-quantum classification model that combines the strengths of CNNs for feature extraction and VQCs as the classifier, has been proposed. For feature extraction, several CNN architectures such as ResNet, VGG, and DenseNet were explored from which ResNet was chosen for its versatility and superior performance. A pre-trained ResNet on ImageNet is then modified by adding two additional layers to generate a single feature vector, which serves as the input to the VQC. The VQC in ResQ comprises a single qubit, equipped with a Hadamard gate and a Pauli-Y rotation gate, designed for efficient quantum computation. The model's performance is rigorously evaluated on two publicly available datasets: BreakHis and BI. On the BreakHis dataset, ResQ achieves remarkable average accuracies of 99.83%, 99.68%, 99.59%, and 98.49% for images at 40×, 100×, 200×, and 400× magnifications, respectively. For the smaller BI dataset, the model achieves a competitive accuracy of 88.96%. These results have been benchmarked against classical counterparts and recent state-of-the-art networks, demonstrating the potential of ResQ as a powerful hybrid model for breast cancer classification. Additionally, a detailed circuit-level analysis of ResQ is also conducted and compared with other hybrid classical-quantum models. Executing the identified best ResQ architecture on a noisy real quantum processor is also subsumed in this work. This extensive analysis shows that ResQ is not only efficient but also compact, which is a significant advantage when using limited quantum resources. These features make ResQ a practical option for medical image classification, especially in environments where computational resources are constrained.

In summary:

- The proposed architecture, ResQ, combines the strengths of CNNs for feature extraction and VQCs as the classifier.

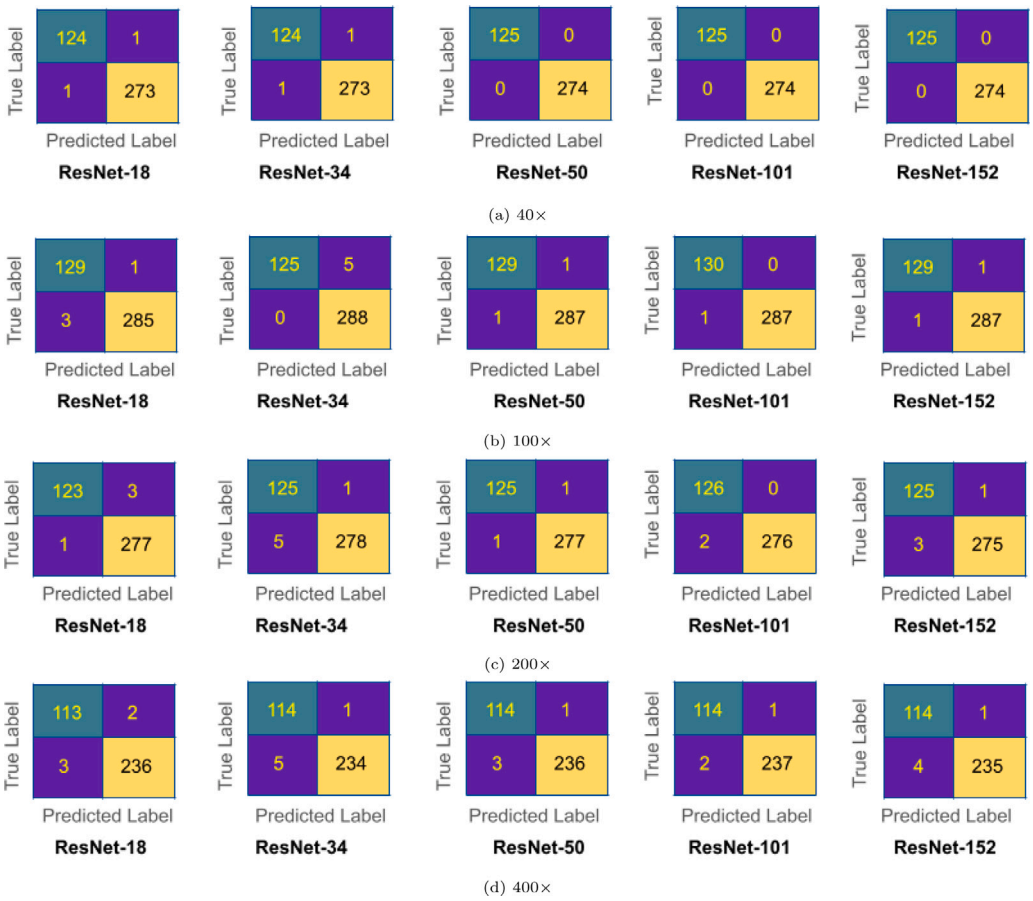


Fig. 12. Confusion matrix for images of the BreakHis test dataset at different magnifications (a) 40× (b) 100× (c) 200× (d) 400× classified using the proposed ResQ network architecture.

- ResNet has been chosen as the pre-trained classical CNN for the hybrid network after considering multiple CNNs like DenseNet-121, DenseNet-201, VGG-16, VGG-19, MobileNet, etc.
- The one-qubit VQC in ResQ significantly reduces quantum resource requirements, making it more feasible on NISQ devices, which are constrained by qubit count and coherence times.
- It has been duly established that the proposed architecture is scalable to multi-qubit versions — a requirement for more complex or imbalanced datasets.
- Performance of ResQ is rigorously evaluated on two publicly available datasets: BreakHis and Bioimaging, achieving competent average accuracies replete with extensive post-hoc statistical tests, in classifying benign and malignant tumors.
- For the VQC execution, both the quantum simulator and noisy quantum hardware have been used.
- Efficiency as well as compactness of ResQ is detailed through a circuit-level analysis coupled with comparison with other hybrid classical-quantum models.
- Profiling breakdown of training time with respect to classical processing (forward pass and backward pass), and processing in the VQC (parameter-shift rule) have been showcased.
- Analysis of both actual training execution times as well as theoretical time complexity has been done.
- Classical counterparts performing similar tasks of classifications and recent state-of-the-art networks have been studied thoroughly, demonstrating the potential of ResQ as a powerful hybrid model for breast cancer classification.

Despite the progress, this work is subject to several limitations stemming from the constraints of current quantum hardware. Quantum devices are inherently prone to noise and errors due to the decoherence of qubits, which diminishes their performance. Additionally, restricted access quotas and lengthy queues for real quantum machines further impede the ability to thoroughly investigate the scalability of ResQ. Experimental results also indicate that bias in dataset partitioning can negatively impact the model's performance. For non-homogeneity in patient-split, the model fails to provide quantitative metric values that are as good as those for the uniform or random split. Thus, the inclination of the model towards dataset bias cannot be ruled out. Quantum technologies are relatively new, hence require comprehensive validation and compliance approvals before successful integration in clinical environments. Also, the proposed network remains to be explored for medical tasks demanding multi-level classification. Future work could be aimed at exploring more advanced quantum circuit designs. Quantum optimization algorithms, too, can be used to examine the impact on efficiency. This could extend the utility of ResQ beyond classification, broadening its potential impact in medical imaging and other domains. Given that quantum technologies are still emerging, they require extensive validation and regulatory approval before they can be effectively integrated into clinical practice. Additionally, the proposed network has yet to be evaluated on medical tasks that involve multi-level classification. Future research could focus on investigating more sophisticated quantum circuit architectures. Exploring quantum optimization algorithms may also help assess and enhance the model's efficiency. Such advancements could expand the applicability of ResQ beyond simple classification, increasing its relevance in medical imaging and other fields. Further work can aim to improve the compactness of ResQ and adapt it for additional tasks, such as segmentation and localization. Utilizing multi-qubit systems may also enable more advanced pattern recognition and optimize feature extraction for these complex tasks.

Glossary of quantum computing terms for medical practitioners

Qubit	The basic unit of information in quantum computing. A qubit is capable of being in a state of $\|0\rangle$, $\|1\rangle$, or both simultaneously (superposition).
Superposition	A quantum property allowing qubits to exist in multiple states ($\|0\rangle$ and $\|1\rangle$) at once, unlike classical bits that can assume either 0 or 1 at a time.
Entanglement	A phenomenon where qubits become interconnected such that the state of one instantly affects the state of another.
Quantum gate	Quantum gates are essentially unitary matrices that perform unitary operations to transform the state of qubits, akin to logic gates in classical computing. Examples of single-qubit quantum gates are the Hadamard gate and the Pauli gates.
Quantum circuit	A collection of quantum gates interconnected sequentially and applied to qubits to perform a computation or classification task.
Measurement	The act of observing a qubit's state, collapsing it from a superposition to either 0 or 1.
Variational Quantum Classifier (VQC)	A quantum model with adjustable parameters that are optimized using classical methods for specific tasks.
Parameterized quantum circuit	A quantum circuit containing gates with tunable parameters, trained to perform specific tasks like classification.
Hybrid classical-quantum model	A computational model that integrates classical algorithms with quantum circuits for enhanced performance.
Parameter-shift rule	A technique used to compute gradients in quantum circuits for model training via classical optimization.
Quantum noise	Unintended interference or errors in quantum computations due to hardware limitations.
NISQ (Noisy Intermediate-Scale Quantum)	The current generation of quantum processors with limited qubits and prone to operational noise.
Quantum feature encoding	The method of converting classical input data into quantum states for use in quantum circuits.
Ansatz	A predefined structure of a quantum circuit with tunable parameters, used in variational algorithms.
Barren Plateau	A situation in quantum training where gradients vanish, making learning and optimization difficult.
Quantum speedup	The theoretical or practical advantage of solving problems faster using quantum algorithms than classical ones.

CRediT authorship contribution statement

Dibyasree Guha: Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Somenath Kuiry:** Writing – review & editing, Methodology, Validation, Formal analysis. **Shyamali Mitra:** Validation, Investigation. **Siddhartha Bhattacharyya:** Validation, Supervision. **Nibaran Das:** Validation, Supervision, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used generative AI in order **to improve the readability and language of the manuscript**. After using this tool, the authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

A variational quantum circuit Q_C of depth d is represented as

$$Q_C = \mathcal{QL}_1 \cdot \mathcal{QL}_2 \cdot \mathcal{QL}_3 \dots \mathcal{QL}_d. \tag{A.1}$$

This can be written as a collection of unitary operations as

$$U(\theta)|state_{inp}\rangle = U_n(\theta_n)U_{n-1}(\theta_{n-1}) \cdots U_1(\theta_1)|state_{inp}\rangle. \tag{A.2}$$

The output of the variational circuit in this work can be represented as the expectation of an observable $\hat{A}$, produced by a single parameterized gate, with θ as the parameter. Therefore, the quantum function $f(\theta)$ can be depicted by

$$f(\theta) = \langle 0|U^{\dagger}(\theta)\hat{A}U(\theta)|0\rangle, \tag{A.3}$$

where, $U^{\dagger}(\theta)$ is the complex conjugate of $U(\theta)$. It can be easily observed that the unitary conjugation $U^{\dagger}(\theta)\hat{A}U(\theta)$ is representable as a linear transformation $\mathcal{X}$ over the observable $\hat{A}$, i.e.,

$$U^{\dagger}(\theta)\hat{A}U(\theta) = \mathcal{X}_{\theta}(\hat{A}). \tag{A.4}$$

This leads to the fact that a well-defined gradient for the output quantum function $f(\theta)$ exists, as the linear transformation has been proved to depend solely on the parameter θ. The gradient definition is given by

$$\frac{d}{d\theta} f(state_{inp}; \theta) = \langle state_{inp}| \frac{d}{d\theta} \mathcal{X}_{\theta}(\hat{A})|state_{inp}\rangle, \tag{A.5}$$

which is evidently a real number. This provides a useful insight that the same transformation $\mathcal{X}$ or a linear combination of it, equipped with a different set of parameters, may be put to use to express gradients in various scenarios. The gradient in many cases can thus be represented as

$$\frac{d}{d\theta} \mathcal{X}_{\theta}(\hat{A}) = r\left[\mathcal{X}_{\theta+s}(\hat{A}) - \mathcal{X}_{\theta-s}(\hat{A})\right], \tag{A.6}$$

where, r and s respectively denote a multiplier constant and a $shift$ on the parameter θ, both of which are independent of the value of θ and depend on the type of the transformation $\mathcal{X}$. Now, any parameterized gate, being a unitary, can be expressed in terms of a Hermitian operator and the gate parameter. Assuming α as the gate parameter and H as the Hermitian operator

$$U(\alpha) = exp(i\alpha H). \tag{A.7}$$

Considering a unitary parameterized gate of the form

$$U(\theta) = exp\left(-i\frac{\theta}{2}\hat{P}\right), \tag{A.8}$$

where, $\hat{P} = \hat{P}^{\dagger}$ is a Pauli operator, the gradient is given by

$$\frac{d}{d\theta}U(\theta) = -\frac{i}{2}\hat{P}U(\theta) = -\frac{i}{2}U(\theta)\hat{P}. \tag{A.9}$$

Combining Eqs. (A.4) and (A.5) and substituting from Eq. (A.9),

$$\begin{aligned}\frac{d}{d\theta}f(state_{inp};\theta) &= \frac{i}{2}\langle state_{inp}|U^{\dagger}(\theta)\left(P\hat{A} - \hat{A}P\right)U(\theta)|state_{inp}\rangle \\ &= \frac{i}{2}\langle state_{inp}|U^{\dagger}(\theta)\left[P,\hat{A}\right]U(\theta)|state_{inp}\rangle,\end{aligned} \tag{A.10}$$

where, $\left[P,\hat{A}\right] = P\hat{A} - \hat{A}P$ is the commutator of matrices P and $\hat{A}$. Using the mathematical identity for commutators on Pauli operators [62]

$$\left[P,\hat{A}\right] = -i\left(U^{\dagger}\left(\frac{\pi}{2}\right)\hat{A}U\left(\frac{\pi}{2}\right) - U^{\dagger}\left(-\frac{\pi}{2}\right)\hat{A}U\left(-\frac{\pi}{2}\right)\right), \tag{A.11}$$

The gradient expression becomes

$$\begin{aligned}\frac{d}{d\theta}f(state_{inp};\theta) = &\frac{1}{2}\langle state_{inp}|U^{\dagger}(\theta+\frac{\pi}{2})\hat{A}U(\theta+\frac{\pi}{2})|state_{inp}\rangle \\ &-\frac{1}{2}\langle state_{inp}|U^{\dagger}(\theta-\frac{\pi}{2})\hat{A}U(\theta-\frac{\pi}{2})|state_{inp}\rangle.\end{aligned} \tag{A.12}$$

Thus, the hybrid model is trained using the parameter-shift rule for gradient evaluation as

$$\frac{d}{d\theta}f(\theta) = r\left[f\left(\theta+\frac{\pi}{4r}\right) - f\left(\theta-\frac{\pi}{4r}\right)\right], \tag{A.13}$$

where, $r = \frac{1}{2}$ for a single qubit Pauli rotation gate.

Data availability

The data that have been used in this article are publicly available and have been properly cited in the paper.

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